Set contrasts

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Every gene set from `04_Gene_Sets`, tested in each of the nine contrasts by two methods that ask different questions:

- fgsea (fast gene set enrichment analysis) is competitive: it ranks every protein by its moderated t from step 02 and asks whether a set's proteins sit higher or lower in that ranking than random sets of the same size. It treats proteins as exchangeable, which they are not (proteins in one complex move together); correlation between members inflates false positives in competitive tests (Wu and Smyth 2012).
- fry (limma) is self-contained: it asks whether the set's proteins move together in the data, rotating residuals under the design, person block and within-person correlation. People are the unit, so at 8 per group it calls little.

Each method's own Benjamini-Hochberg correction runs within one contrast over every set.

Packages

```
library(here)
library(limma)
library(dplyr)
library(tidyr)
library(purrr)
library(tibble)
library(ggplot2)
library(grid)
library(ComplexHeatmap)
source(here("00_Setup", "Shared_functions", "analysis_reporting_helpers.R"))
params$grouping
```

```
[1] "Muscle_size"
```

Inputs

The protein fit from step 02, this grouping's design, the gene sets, limpa's complete matrix (fry needs one without gaps), the set scores for the overview, and the phenotype for the trait tracks.

```
data_dir <- file.path("03_Group_comparisons", params$grouping, "c_data")
inputs <- c(
  gene_sets = "01_Processing/04_Gene_Sets/c_data/gene_sets.rds",
  fit = file.path(data_dir, "02_fit.rds"),
  design = file.path(data_dir, "01_design.rds"),
  limpa = "01_Processing/03_Quantification/c_data/DAList_limpa.rds",
  singscore = "01_Processing/05_Set_Scores/c_data/singscore.rds",
  phenotype = "00_Setup/Input/phenotype.csv"
)
paths <- map_chr(inputs, here)
stopifnot(file.exists(paths))
gs <- readRDS(paths[["gene_sets"]])
fit <- readRDS(paths[["fit"]])
d <- readRDS(paths[["design"]])
complete <- subset_for_design(readRDS(paths[["limpa"]]), d)
sets <- gs$sets
protein_map <- gs$protein_map
protein_ids <- protein_map$uniprot_id
eb <- fit$eBayes_fit
contrast_names <- colnames(fit$design$contrast_matrix)
design <- fit$design$design_matrix
subject <- fit$metadata$subject
```

A stale fit would still line up by count, so these checks compare names, not lengths.

```
stopifnot(
  identical(rownames(eb$coefficients), protein_ids),
  identical(colnames(eb$coefficients), contrast_names),
  identical(rownames(complete$data), protein_ids),
  identical(colnames(complete$data), rownames(design)),
  identical(fit$metadata$sample_id, rownames(design)),
  d$baseline_reference %in% contrast_names
)
c(sets = length(sets), proteins = length(protein_ids), runs = ncol(complete$data))
```

```
    sets proteins     runs
    1378     1900       45
```

Protein results from the saved fit

`topTable()` rebuilds the nine contrasts from the saved fit, with the same BH within contrast. Π = p^|logFC| marks the proteins the figure steps draw.

```
protein_results <- map(set_names(contrast_names), function(contrast) {
  topTable(eb, coef = contrast, number = Inf, adjust.method = "BH", sort.by = "none") |>
    rownames_to_column("uniprot_id") |>
    select(uniprot_id, logFC, AveExpr, t, p = P.Value, fdr = adj.P.Val)
}) |>
  list_rbind(names_to = "contrast") |>
```

```
  left_join(protein_map, by = "uniprot_id", relationship = "many-to-one") |>
  mutate(pi_score = p^abs(logFC))
stopifnot(nrow(protein_results) == length(protein_ids) * length(contrast_names))
protein_summary <- protein_results |>
  summarise(
    n_tested = n(), n_up = sum(fdr < 0.05 & logFC > 0), n_down = sum(fdr < 0.05 & logFC < 0),
    min_fdr = signif(min(fdr), 3), .by = contrast
  )
protein_summary
```

```
              contrast n_tested n_up n_down min_fdr
1          Training_HR     1900    0      0  0.7850
2          Training_LR     1900    0      0  0.1760
3             Acute_HR     1900    0      0  0.0508
4             Acute_LR     1900    0      0  0.7960
5       Baseline_HRvLR     1900    0      0  1.0000
6        Trained_HRvLR     1900    0      0  1.0000
7          Acute_HRvLR     1900    0      0  0.4100
8 Training_Interaction     1900    0      0  0.9990
9    Acute_Interaction     1900    0      0  0.9660
```

What each method receives

fgsea gets one ranked vector per contrast: the moderated t of each representative protein, named by gene symbol so its leading edge comes back as genes. fry indexes rows of the matrix, so each set is mapped back to the representative protein `04_Gene_Sets` chose per gene.

```
ranked <- protein_results |>
  filter(selected) |>
  split(~contrast) |>
  map(\(result) set_names(result$t, result$gene))
ranked <- ranked[contrast_names]
gene_map <- filter(protein_map, selected)
set_rows <- map(sets, \(genes) {
  match(gene_map$uniprot_id[match(genes, gene_map$gene)], protein_ids)
})
stopifnot(!any(map_lgl(set_rows, anyNA)))
abundance <- complete$data
correlation <- duplicateCorrelation(abundance, design, block = subject)$consensus.correlation
correlation
```

```
[1] 0.1698778
```

Tests

fgsea estimates p by permutation, so it is seeded. Each package returns its own BH column over the list it was given, kept as `fdr`; none is recomputed. Storey's q is added within contrast and method.

```
set.seed(1)
fgsea_raw <- map(ranked, \(stats) fgsea::fgsea(sets, stats, minSize = 15, maxSize = 500))
set_tests <- bind_rows(
  map(contrast_names, \(contrast) {
    as_tibble(fgsea_raw[[contrast]]) |>
      transmute(
        set_id = pathway, contrast, method = "fgsea", n_proteins = size,
        direction = if_else(NES > 0, "Up", "Down"), nes = NES, p = pval, fdr = padj,
        leading_edge = leadingEdge
      )
  }),
  map(contrast_names, \(contrast) {
    fry(abundance, set_rows, design, fit$design$contrast_matrix[, contrast],
      block = subject, correlation = correlation, sort = "none"
    ) |>
      rownames_to_column("set_id") |>
      transmute(
        set_id, contrast,
        method = "fry", n_proteins = NGenes,
        direction = Direction, nes = NA_real_, p = PValue, fdr = FDR,
        leading_edge = list(NULL)
      )
  })
) |>
  mutate(qvalue = storey_q(p), .by = c(contrast, method))
```

Collapsing overlapping sets

The collections overlap: glycolysis appears in Hallmark, KEGG, Reactome and GO. fgsea's `collapsePathways()` re-tests each significant set conditioned on a more significant set's leading edge and keeps it only if it stays significant. It prunes after testing rather than re-adjusting, so the survivors' FDR is conservative. Survivors are flagged `main`; nothing is dropped.

```
main_sets <- map(set_names(contrast_names), function(contrast) {
  significant <- fgsea_raw[[contrast]][padj < 0.05][order(pval)]
  if (nrow(significant) < 2) {
    return(significant$pathway)
  }
  fgsea::collapsePathways(significant, sets, ranked[[contrast]], pval.threshold =
0.05)$mainPathways
})
main_lookup <- imap(main_sets, \(ids, cn) tibble(contrast = cn, set_id = ids, kept = TRUE)) |>
  list_rbind()
set_tests <- set_tests |>
  left_join(main_lookup, by = c("contrast", "set_id")) |>
  mutate(main = if_else(method == "fgsea", coalesce(kept, FALSE), NA), kept = NULL) |>
  left_join(
    select(gs$set_catalog, set_id, collection, pathway, source_size, description),
    by = "set_id"
```

```
  ) |>
  relocate(contrast, method, set_id, collection, pathway)
```

The baseline reference sorts first, so the reference for how far apart the groups start opens the table.

```
set_summary <- set_tests |>
  summarise(
    n_tested_fry = sum(method == "fry"),
    n_tested_fgsea = sum(method == "fgsea"),
    n_fdr_fgsea = sum(method == "fgsea" & fdr < 0.05),
    n_collapsed = sum(method == "fgsea" & fdr < 0.05 & main),
    n_fdr_fry = sum(method == "fry" & fdr < 0.05),
    .by = contrast
  ) |>
  arrange(contrast != d$baseline_reference)
set_summary
```

```
# A tibble: 9 × 6
  contrast         n_tested_fry n_tested_fgsea n_fdr_fgsea n_collapsed n_fdr_fry
  <chr>                   <int>          <int>       <int>       <int>     <int>
1 Baseline_HRvLR           1378           1378          70          36         0
2 Training_HR              1378           1378         107          34         0
3 Training_LR              1378           1378         116          22         0
4 Acute_HR                 1378           1378         299          83         0
5 Acute_LR                 1378           1378          88          27         0
6 Trained_HRvLR            1378           1378           0           0         0
7 Acute_HRvLR              1378           1378         229          72         0
8 Training_Intera…         1378           1378          64          30         0
9 Acute_Interacti…         1378           1378         206          70         0
```

Sets called, one page per contrast and method

Every fgsea collapse survivor at FDR < 0.05 and every set at fry p < 0.05, 40 sets to a page. `dot_pages()` takes the rows to draw, the x variable and labels, and returns one plot per page.

```
labelled <- set_tests |>
  mutate(
    contrast = factor(contrast, levels = contrast_names),
    label = gsub("\n", " ", enrichVolcano::ev_clean_label(pathway))
  )
dot_pages <- function(rows, x, x_label, method_label) {
  rows |>
    arrange(contrast, .data[[x]]) |>
    mutate(page = (row_number() - 1) %/% 40 + 1, n_pages = max(page), .by = contrast) |>
    group_by(contrast, page) |>
    group_map(\(page_rows, key) {
      ggplot(page_rows, aes(.data[[x]], reorder(label, .data[[x]]), colour = collection)) +
        geom_vline(xintercept = 0, colour = "grey75") +
        geom_point(aes(size = n_proteins)) +
```

```
        scale_colour_brewer(palette = "Dark2", name = NULL) +
        scale_size_continuous(range = c(1.5, 5), name = "proteins") +
        labs(
          title = sprintf("%s, %s", key$contrast, method_label),
          subtitle = sprintf(
            "%d sets. Page %d of %d.",
            sum(rows$contrast == key$contrast), key$page, page_rows$n_pages[1]
          ),
          x = x_label, y = NULL
        ) +
        theme_minimal(base_size = 8) +
        theme(plot.title = element_text(face = "bold"))
    })
}
fgsea_pages <- dot_pages(
  filter(labelled, method == "fgsea", fdr < 0.05, main), "nes",
  "normalised enrichment score", "fgsea collapse survivors at FDR < 0.05"
)
fry_pages <- dot_pages(
  labelled |>
    filter(method == "fry", p < 0.05) |>
    mutate(signed = if_else(direction == "Up", 1, -1) * -log10(p)),
  "signed", "signed -log10 p (positive: up)", "fry sets at p < 0.05"
)
pages <- c(fgsea_pages, fry_pages)
stopifnot(
  sum(map_int(pages, \(g) nrow(g$data))) ==
    nrow(filter(set_tests, method == "fgsea", fdr < 0.05, main)) +
      nrow(filter(set_tests, method == "fry", p < 0.05))
)
length(pages)
```

```
[1] 37
```

```
walk(pages, print)
```

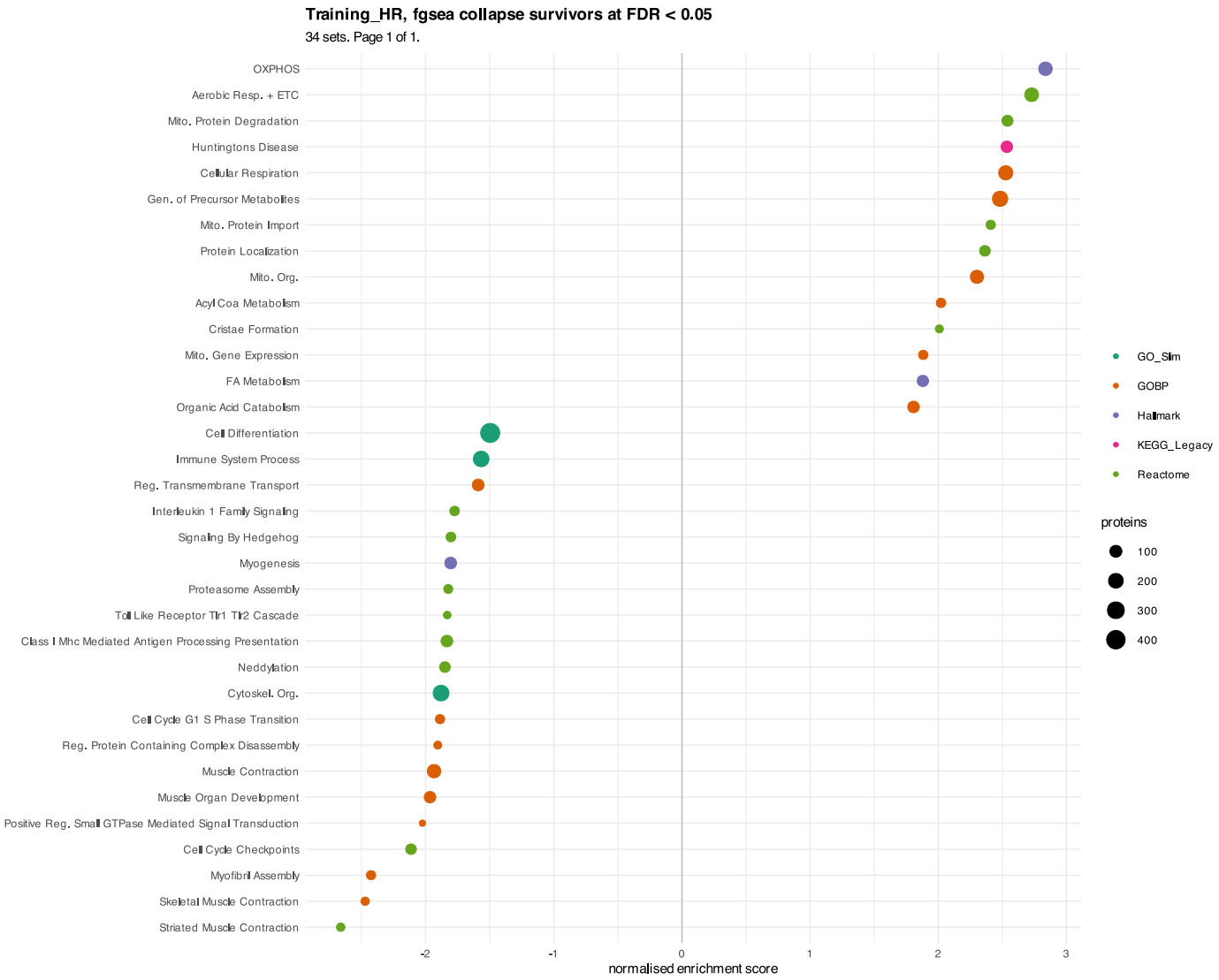


Figure 1: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

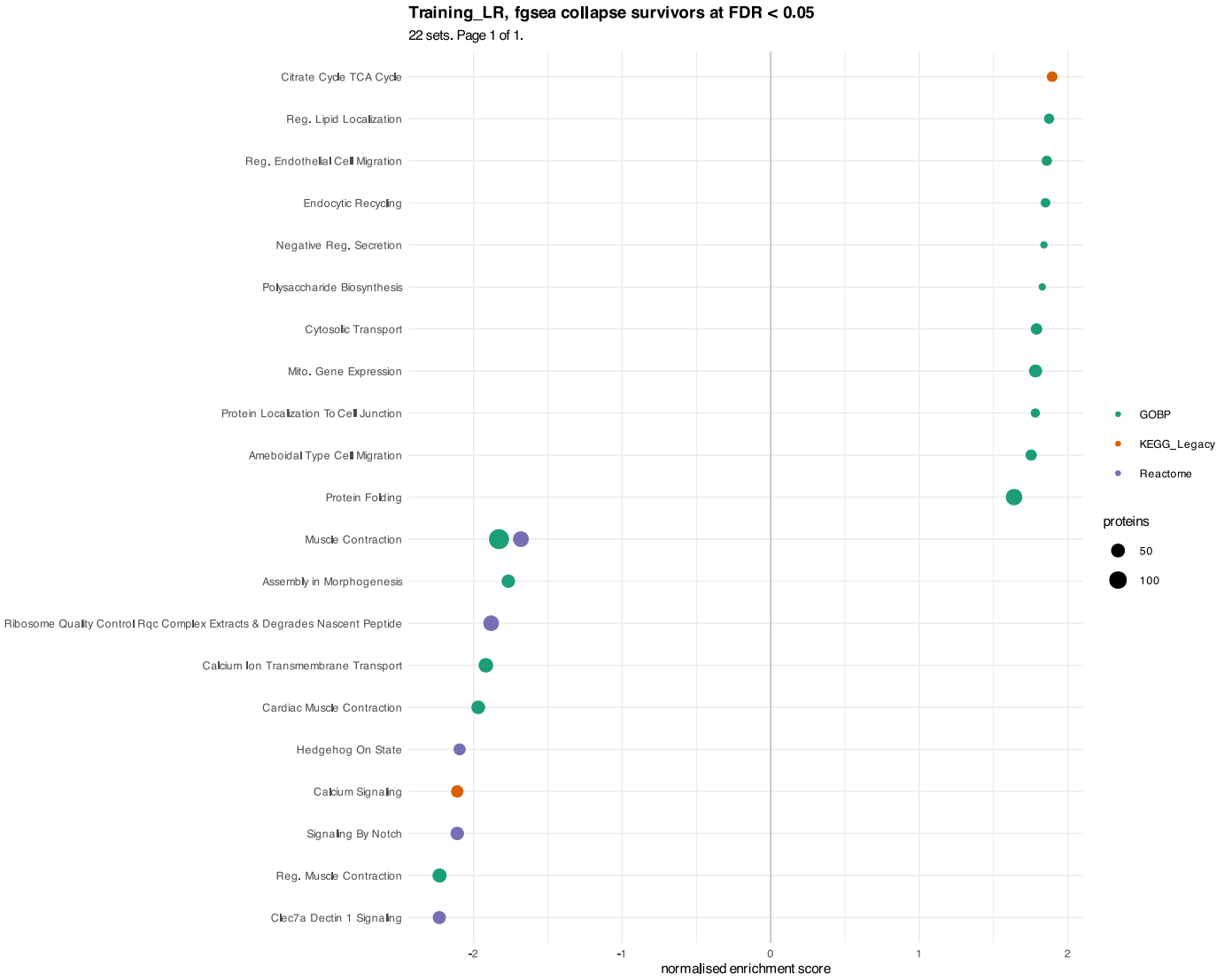


Figure 2: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

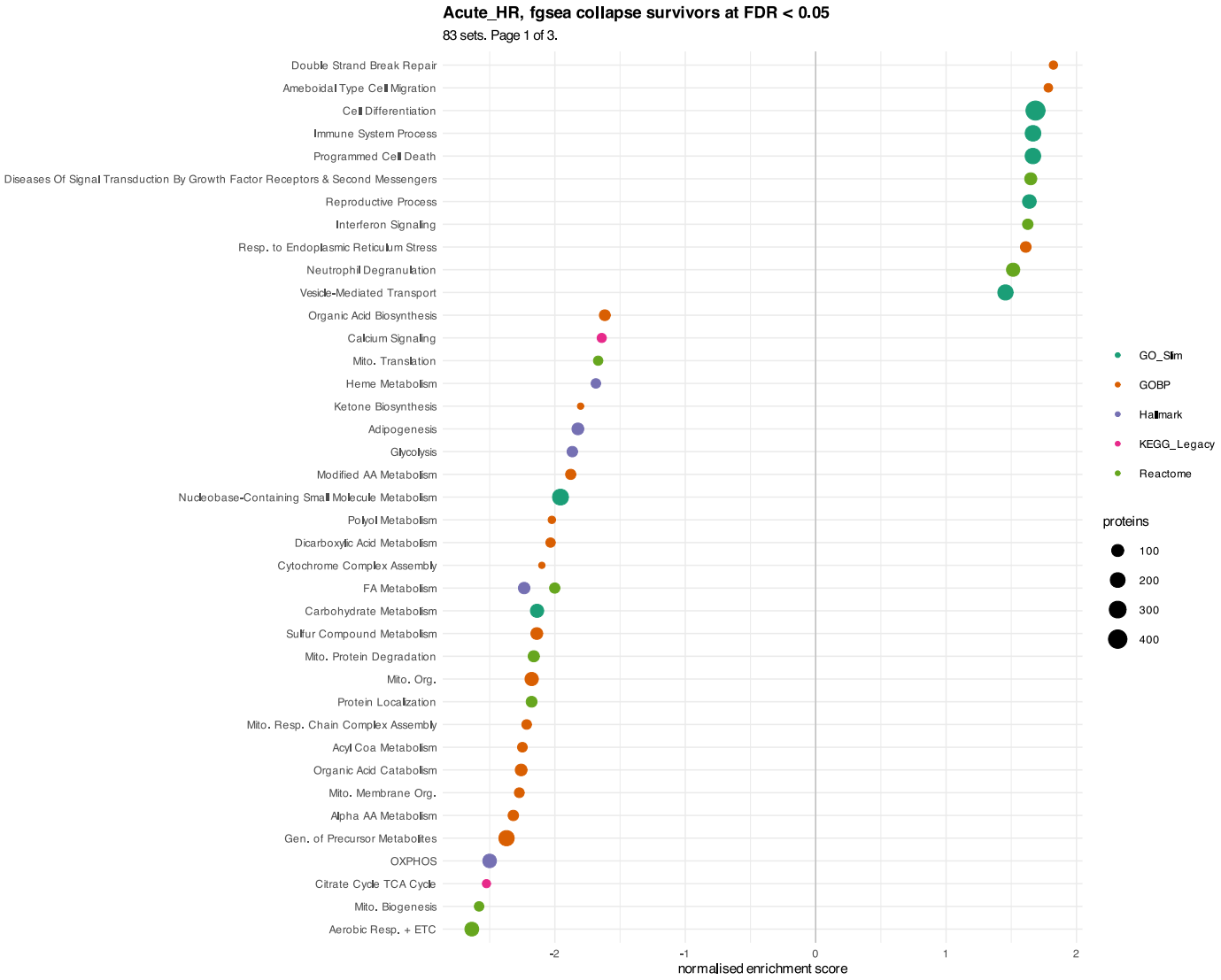


Figure 3: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

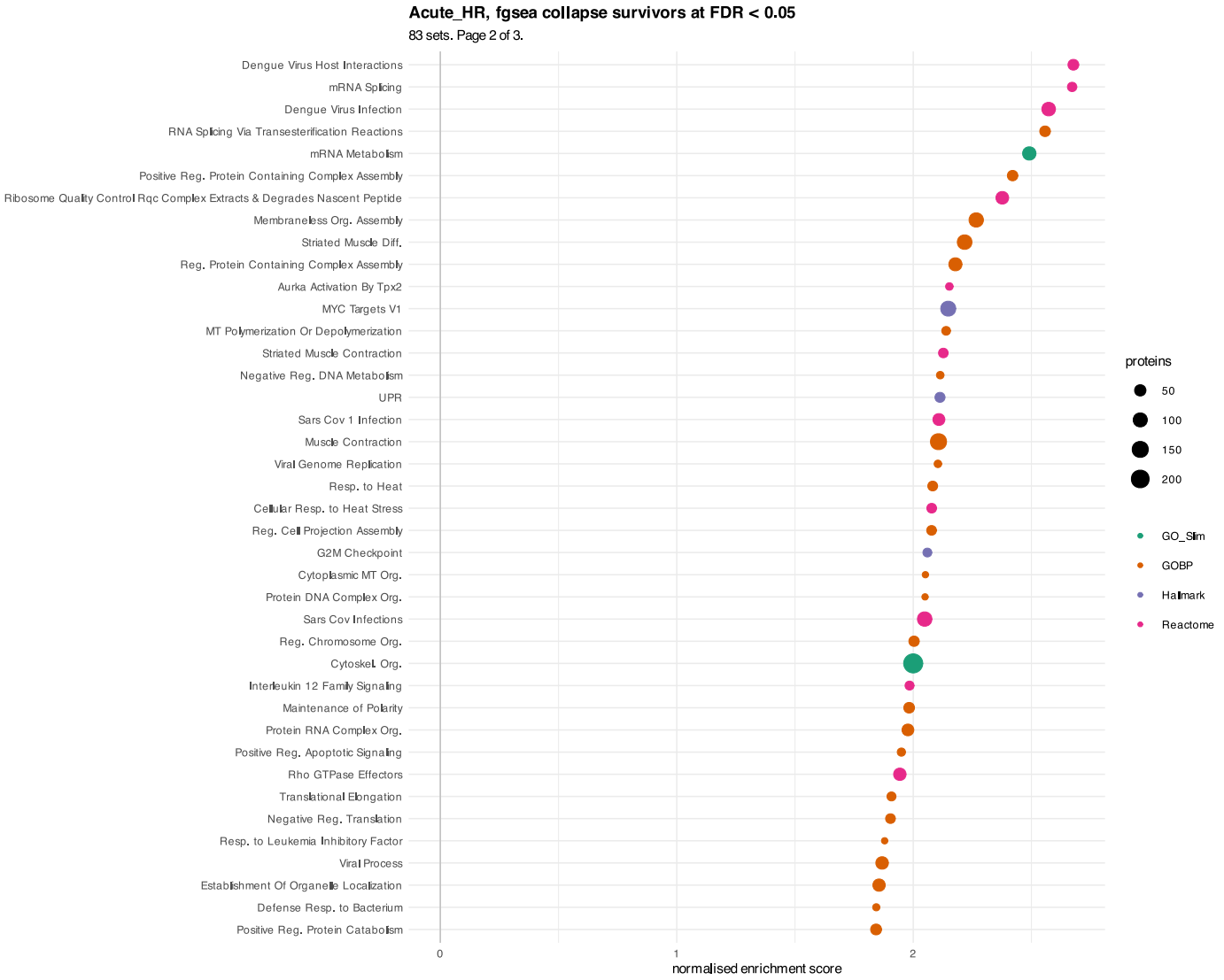


Figure 4: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

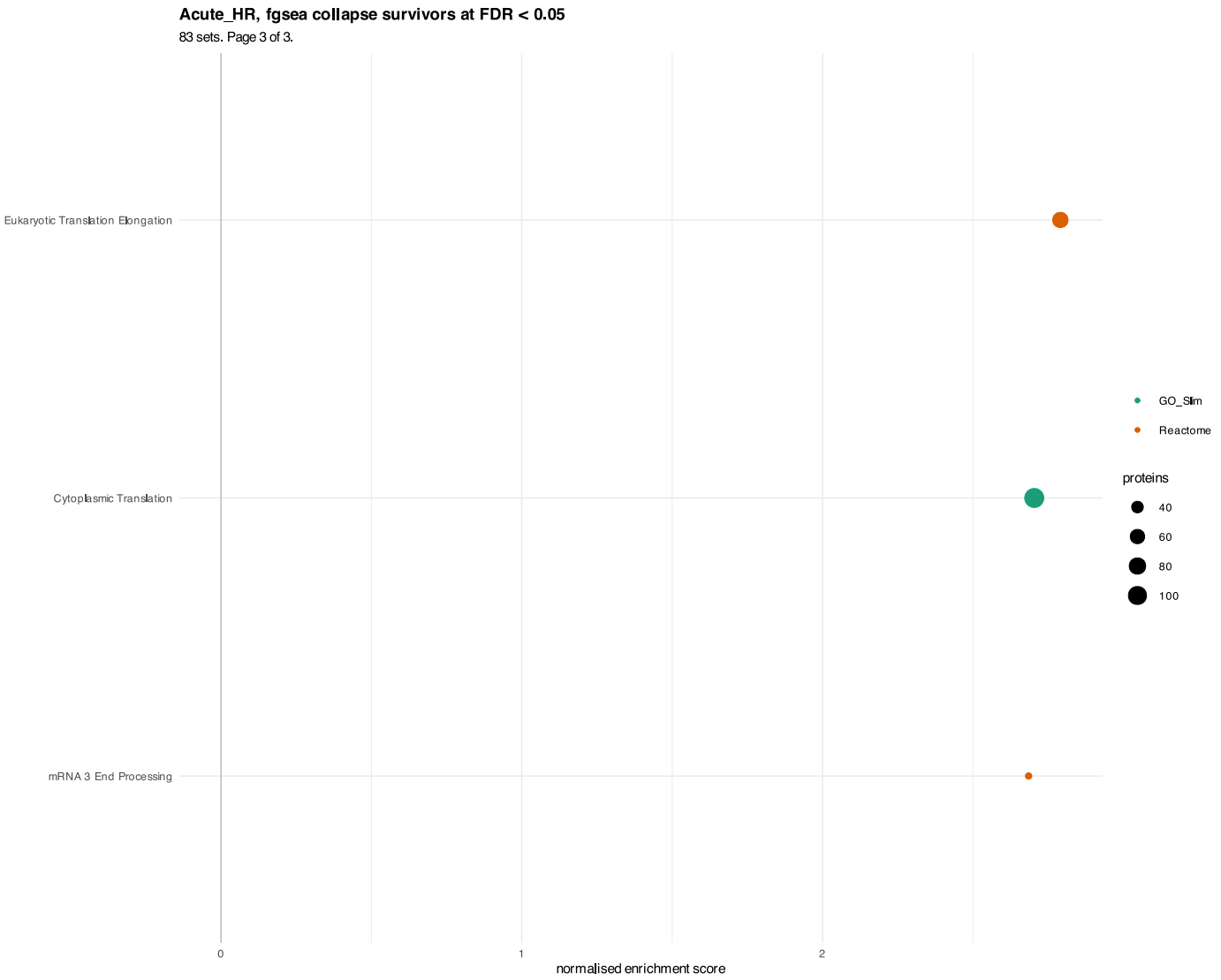


Figure 5: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

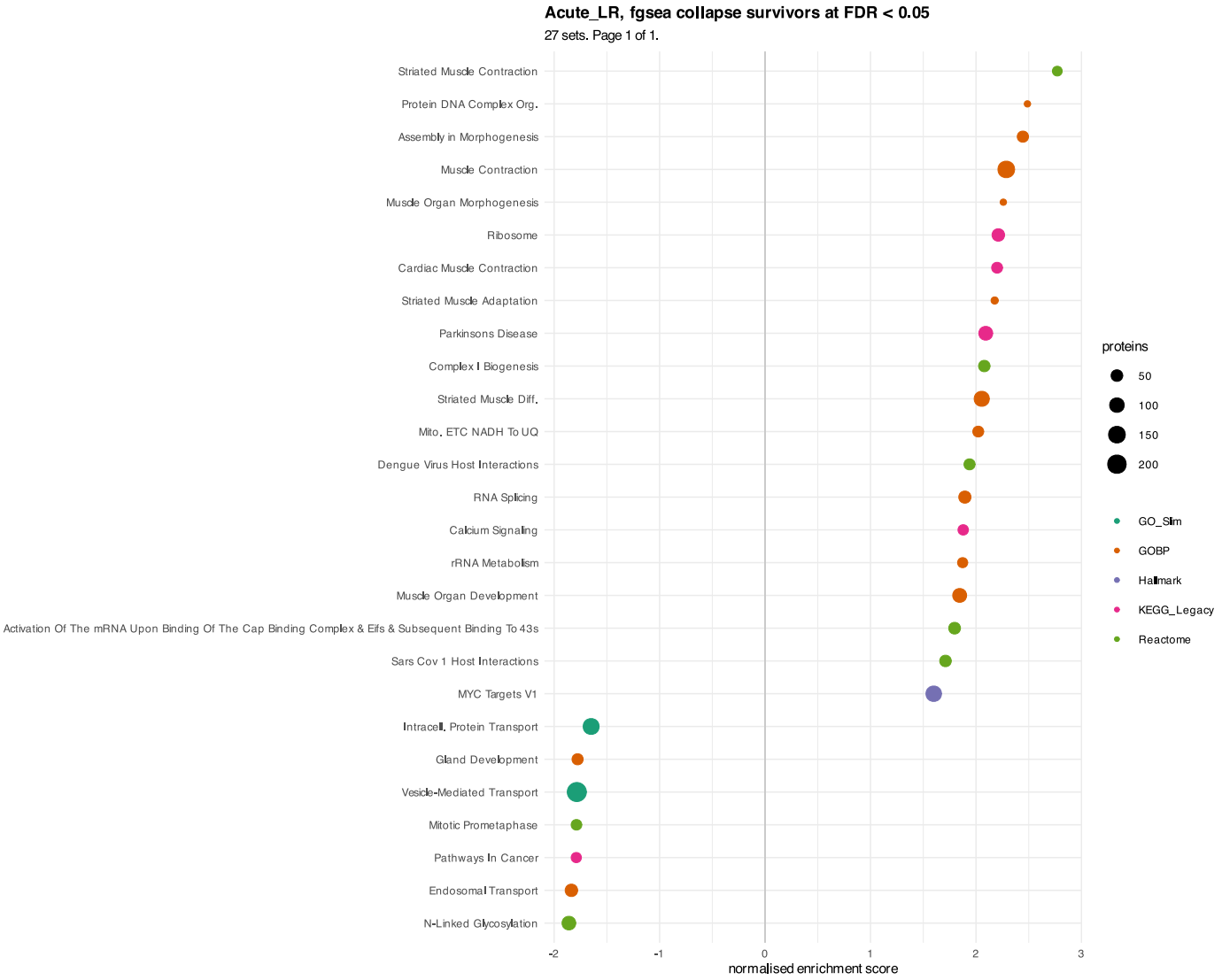


Figure 6: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

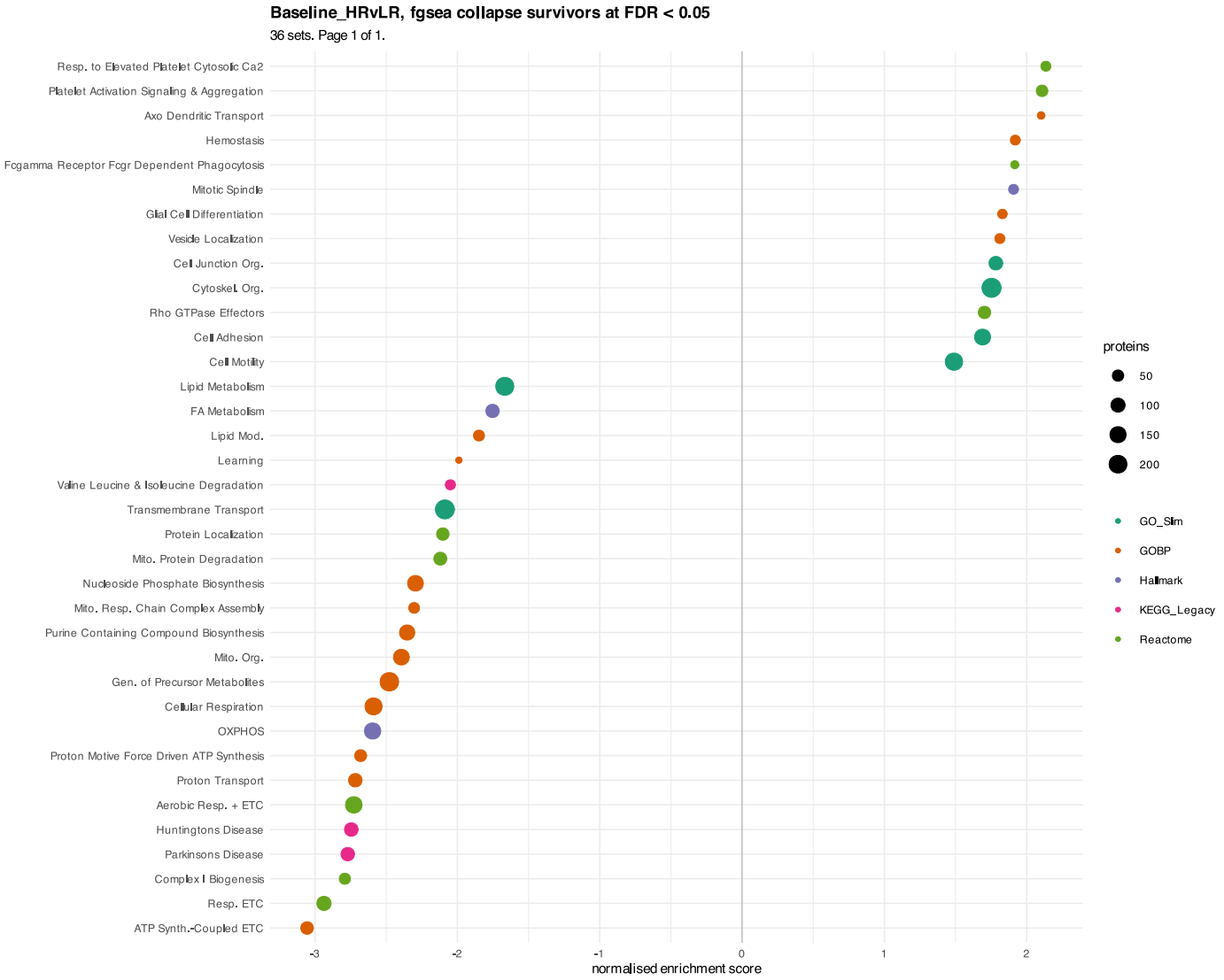


Figure 7: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

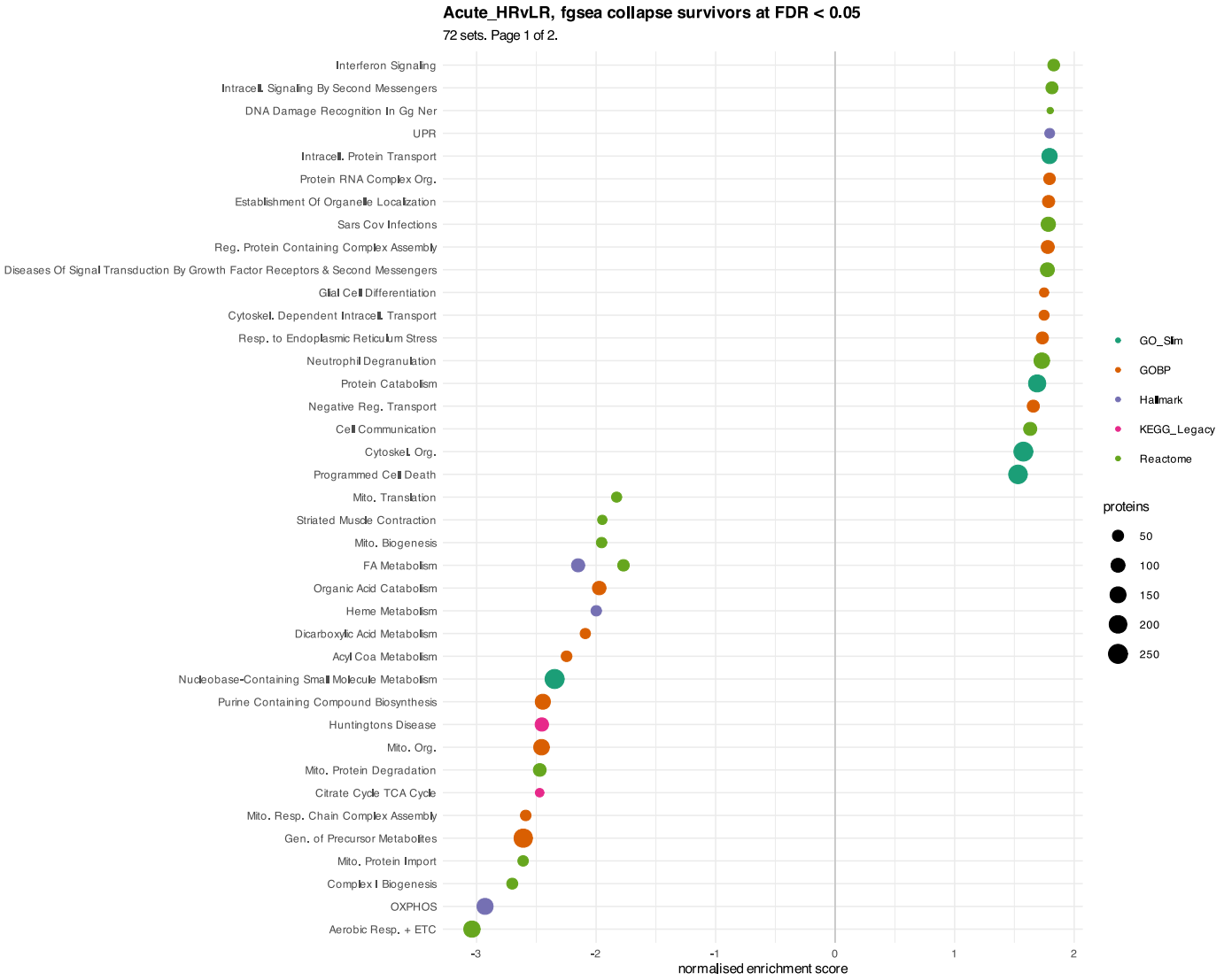


Figure 8: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

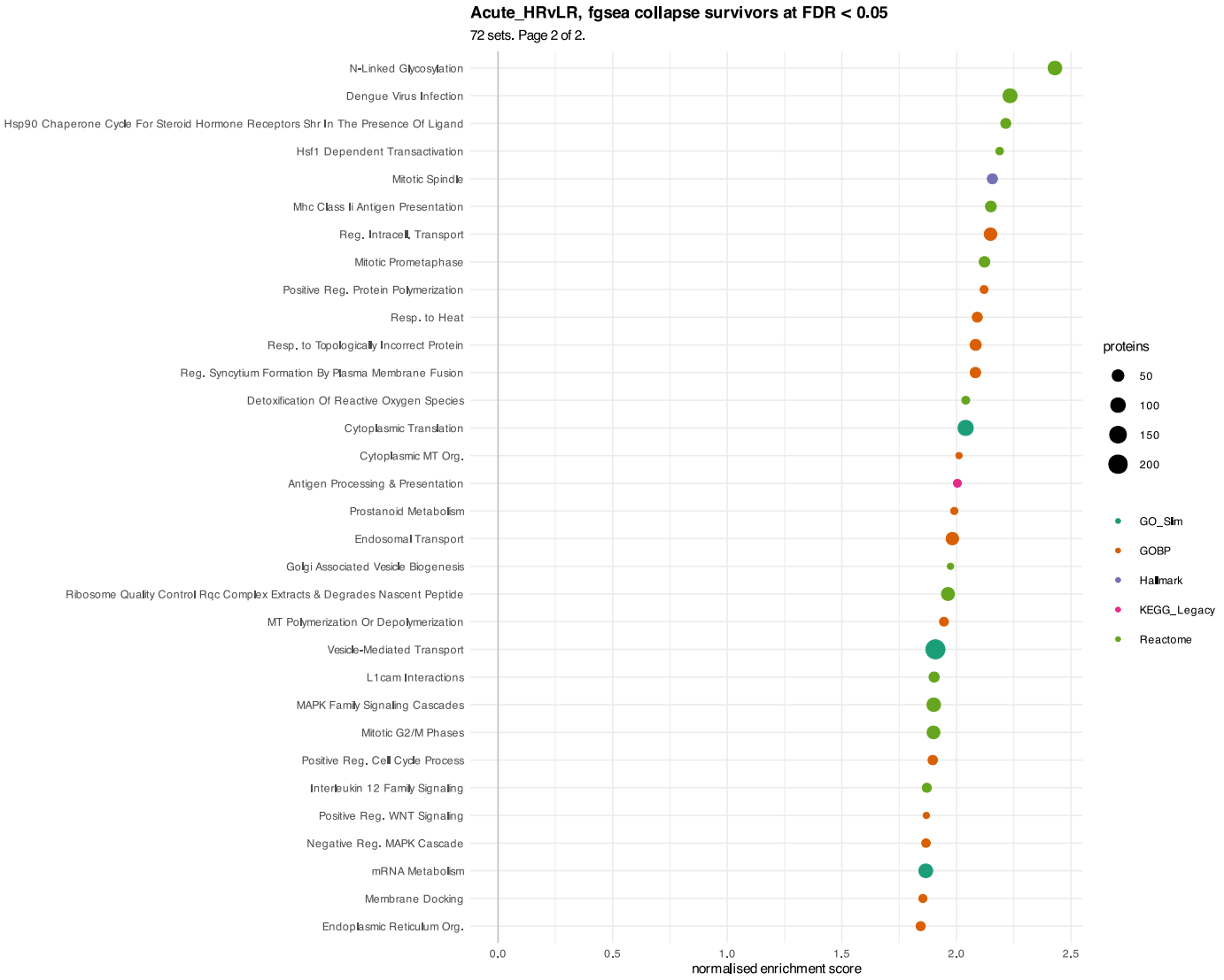


Figure 9: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

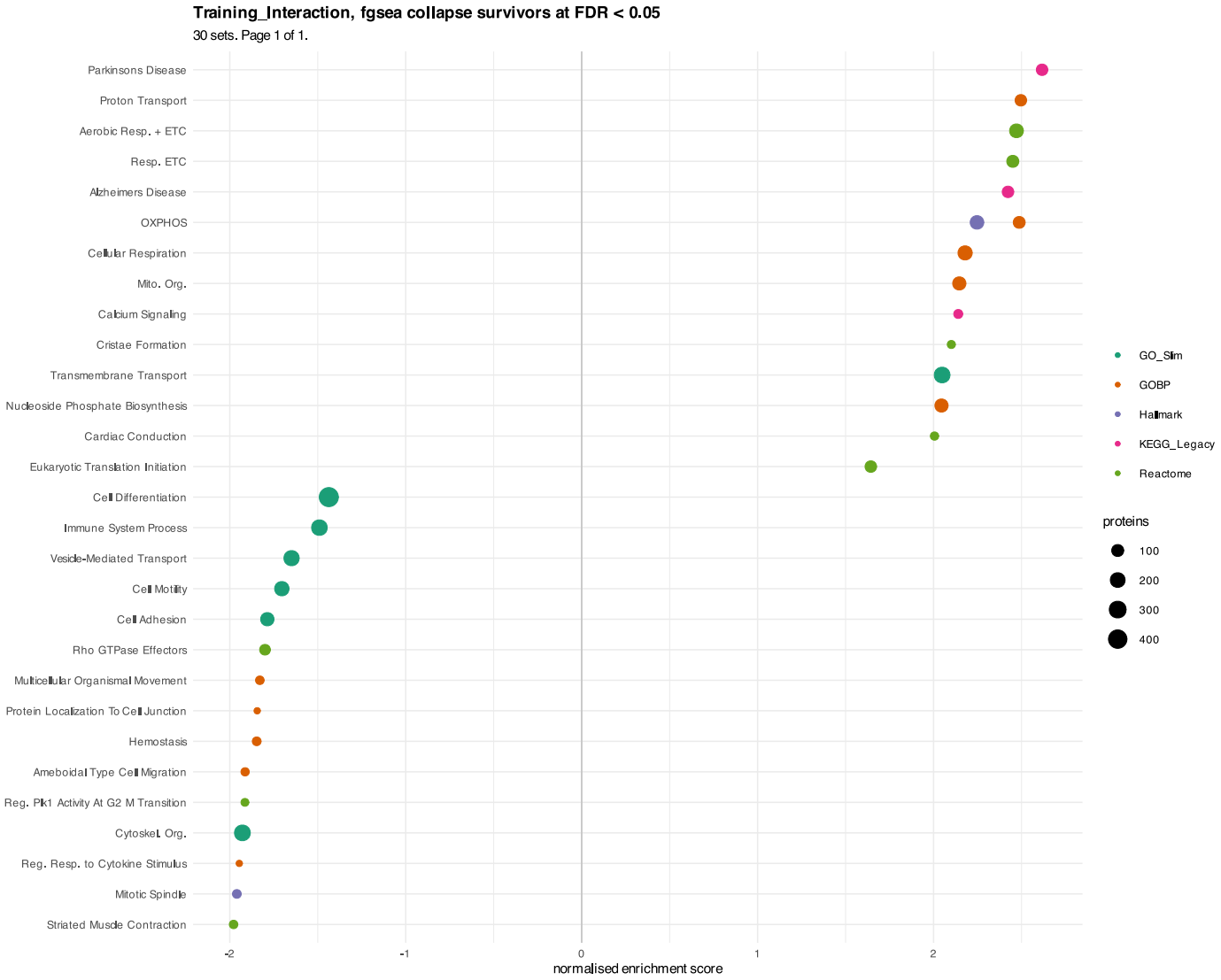


Figure 10: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

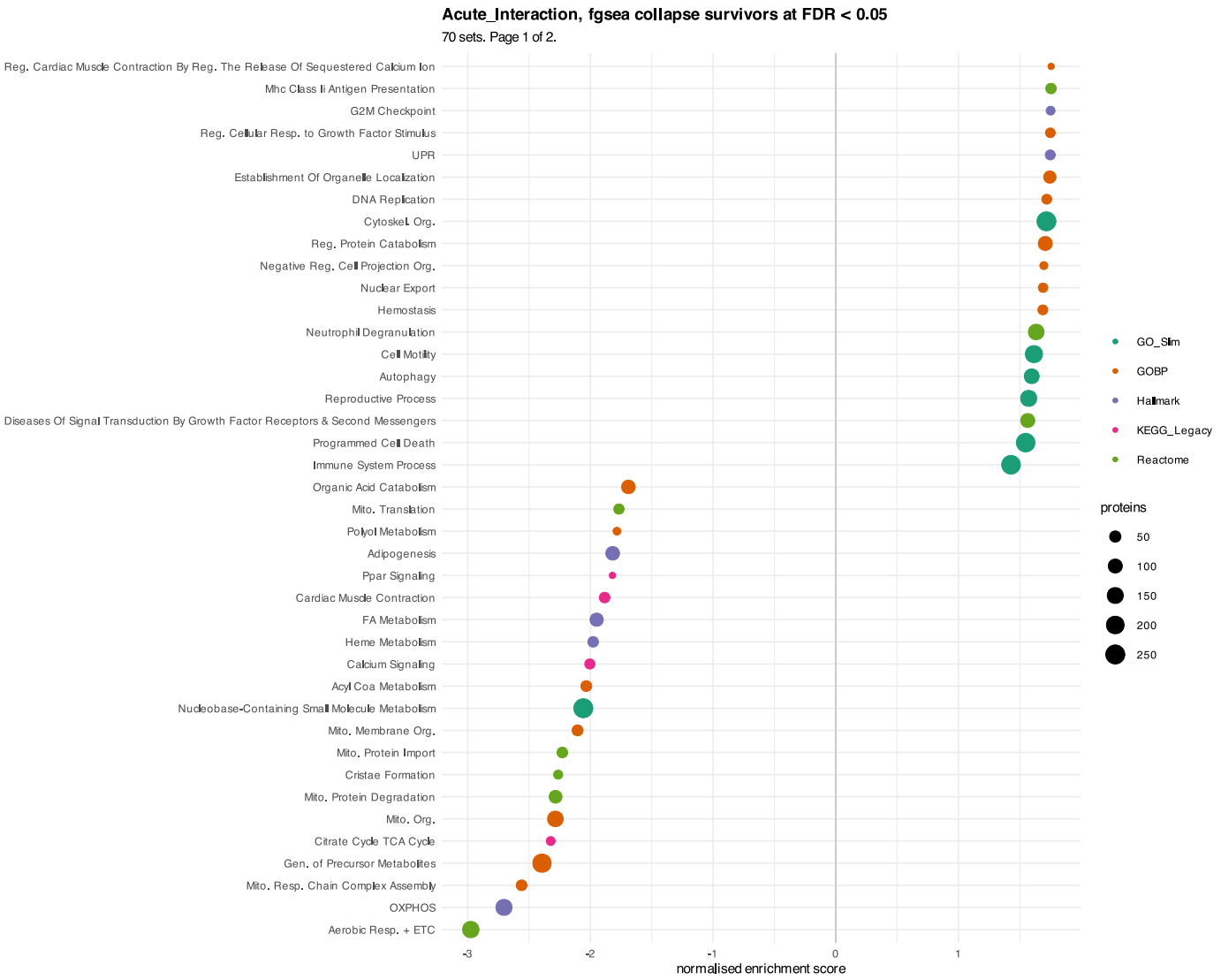


Figure 11: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

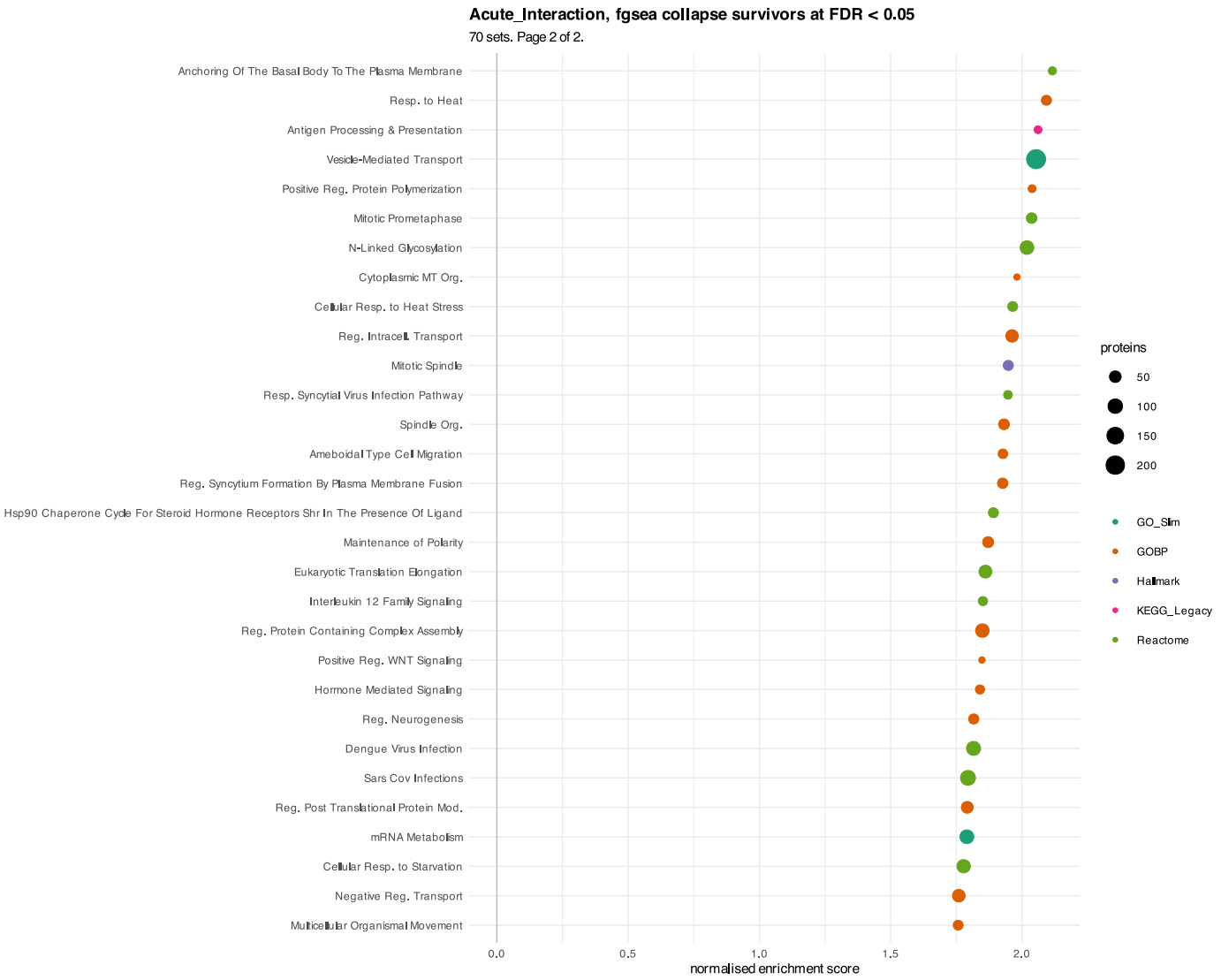


Figure 12: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

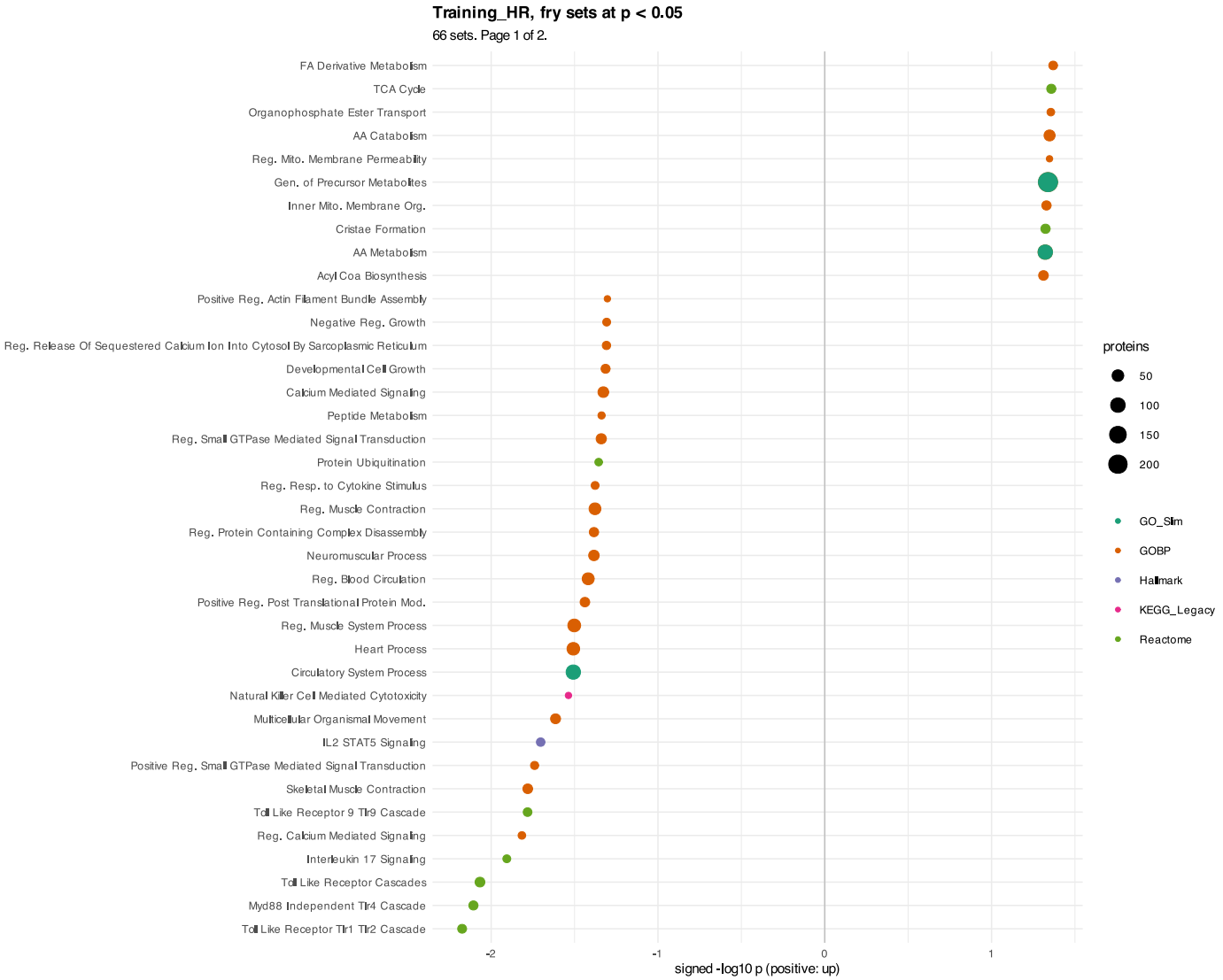


Figure 13: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

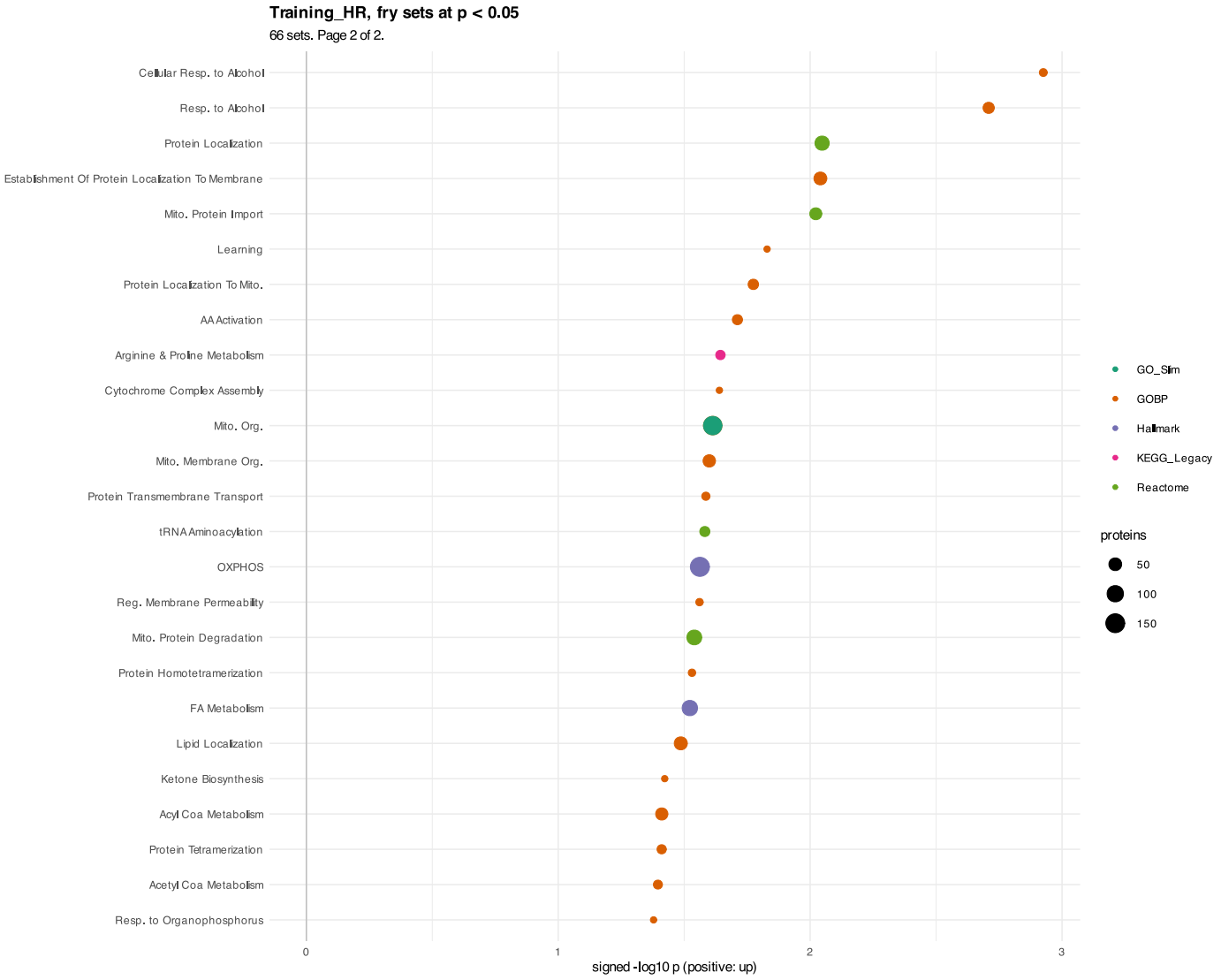


Figure 14: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

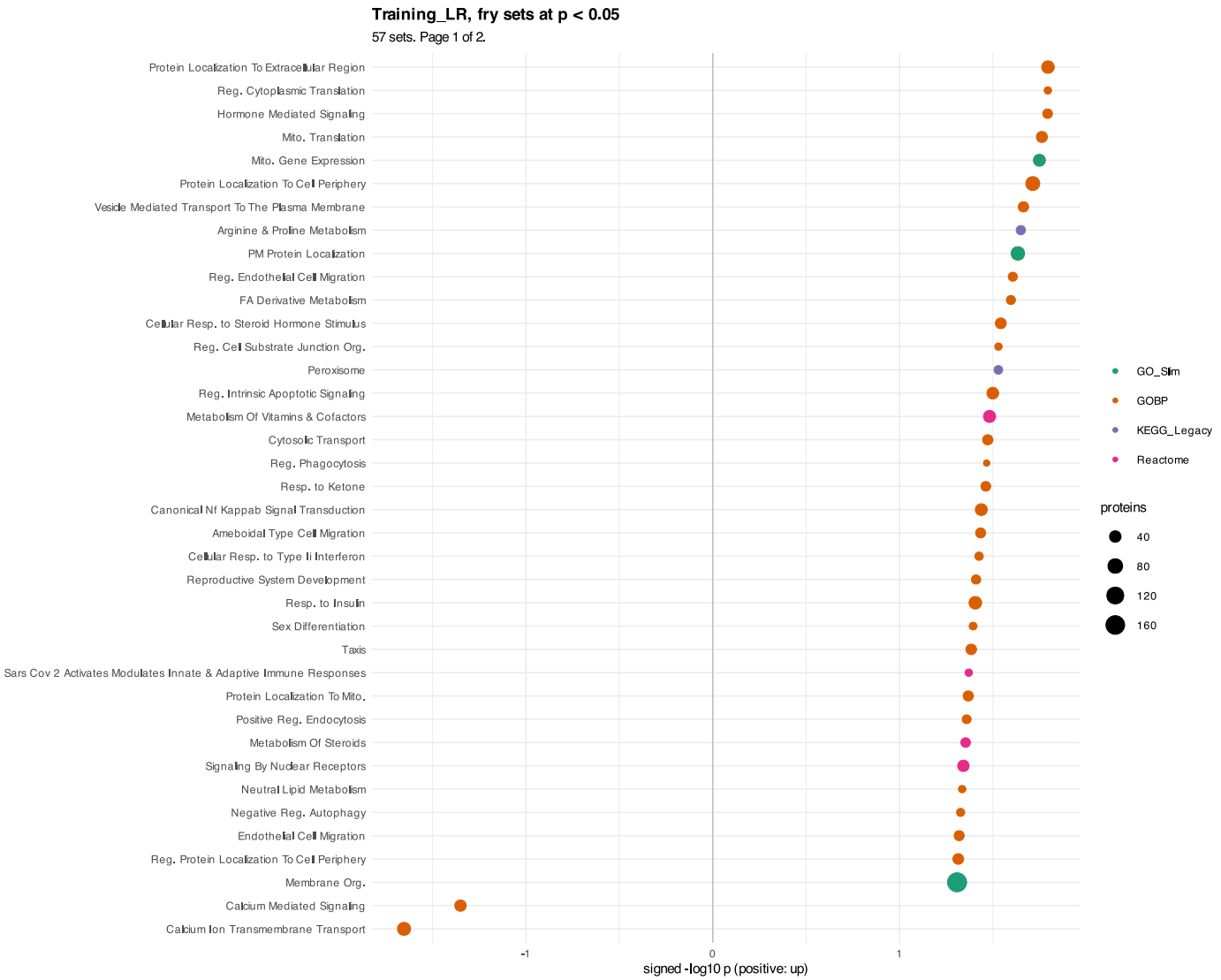


Figure 15: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

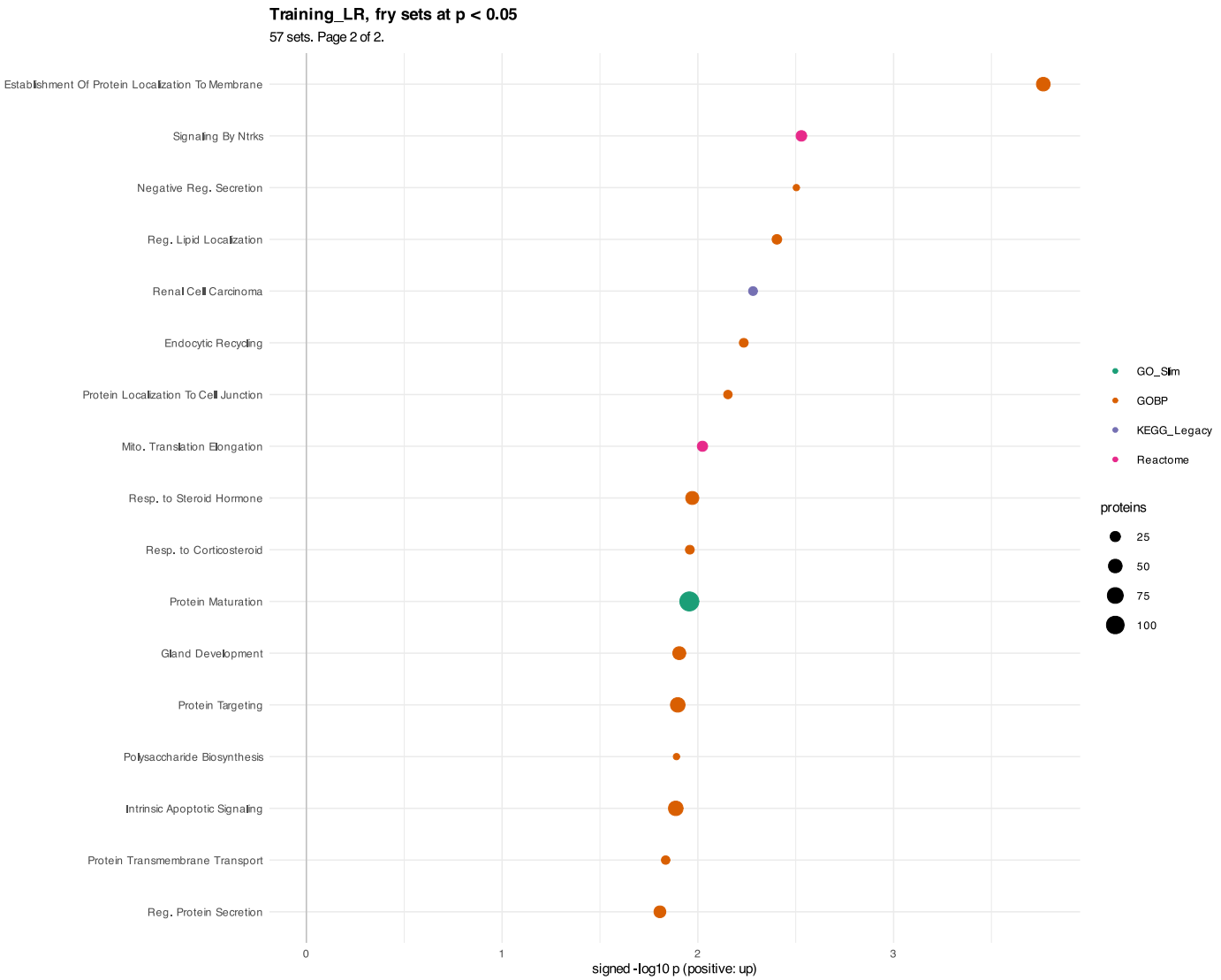


Figure 16: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

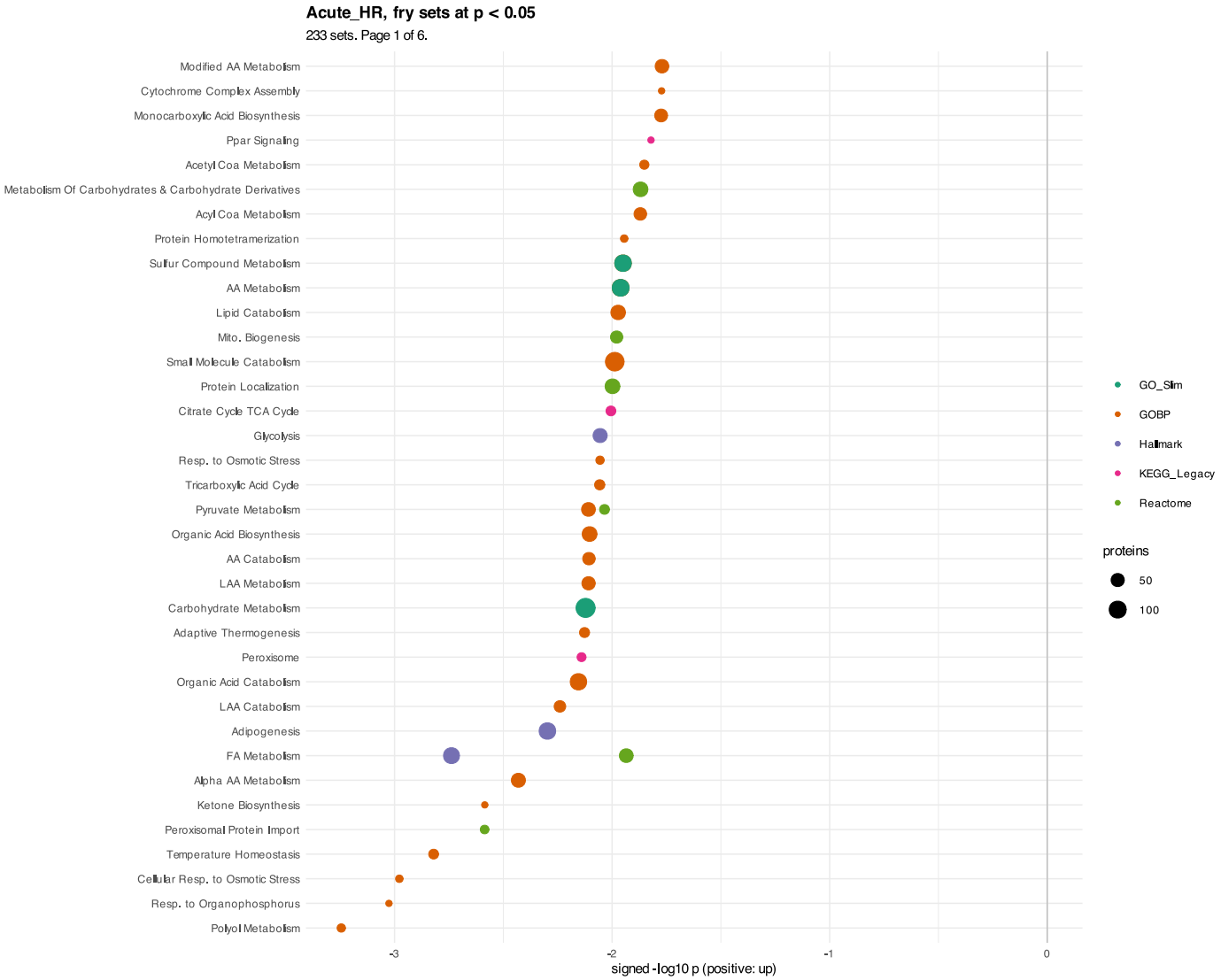


Figure 17: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

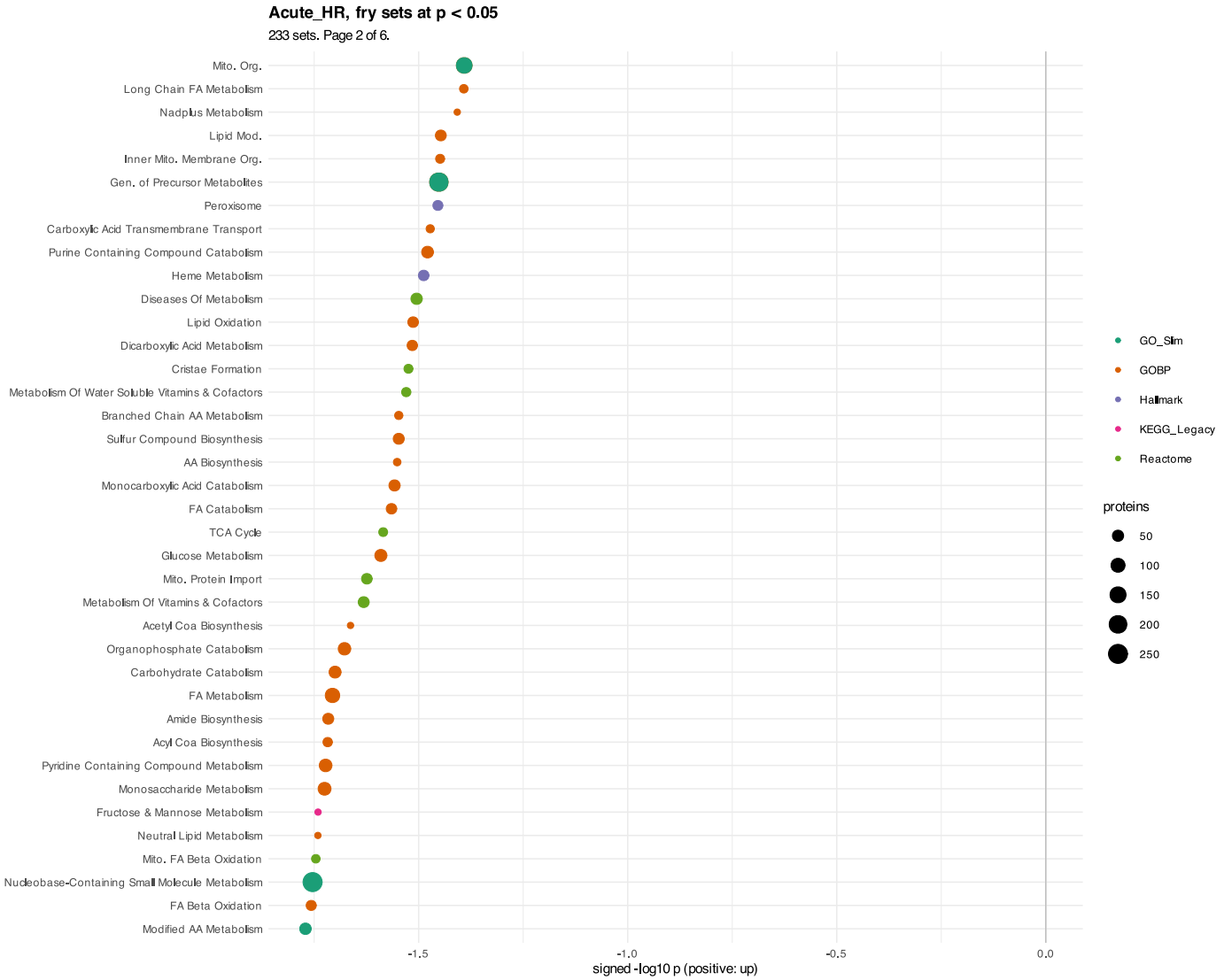


Figure 18: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

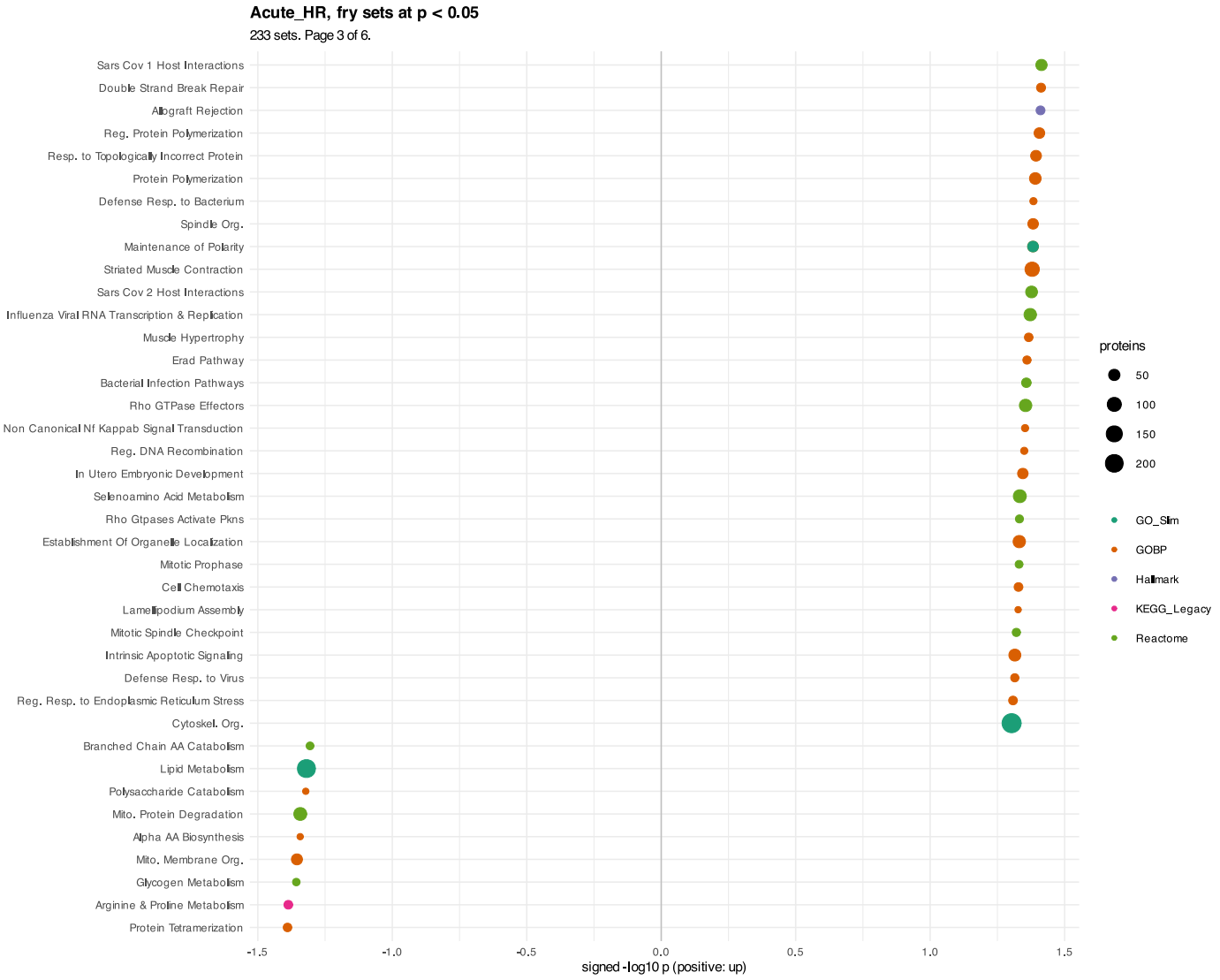


Figure 19: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

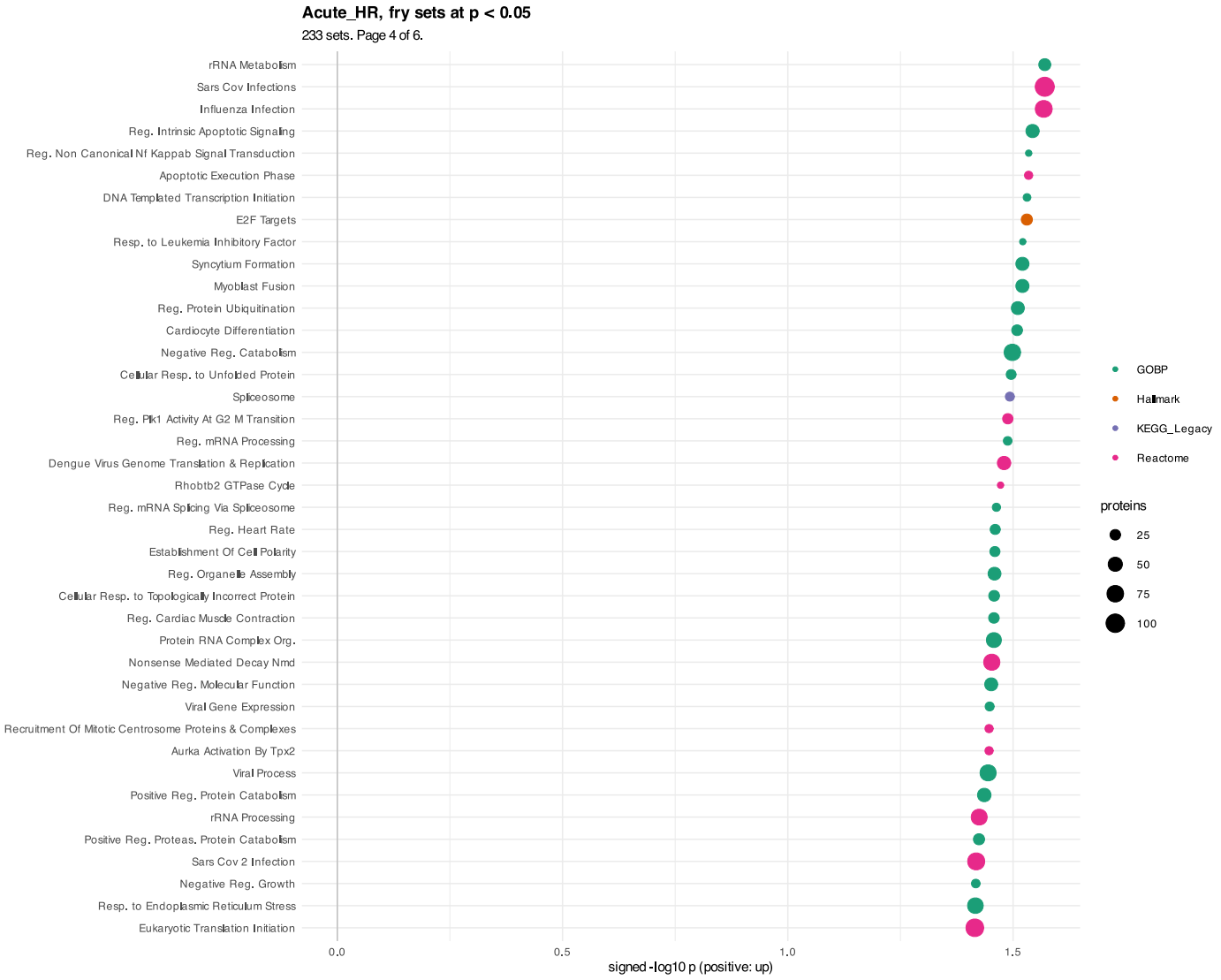


Figure 20: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

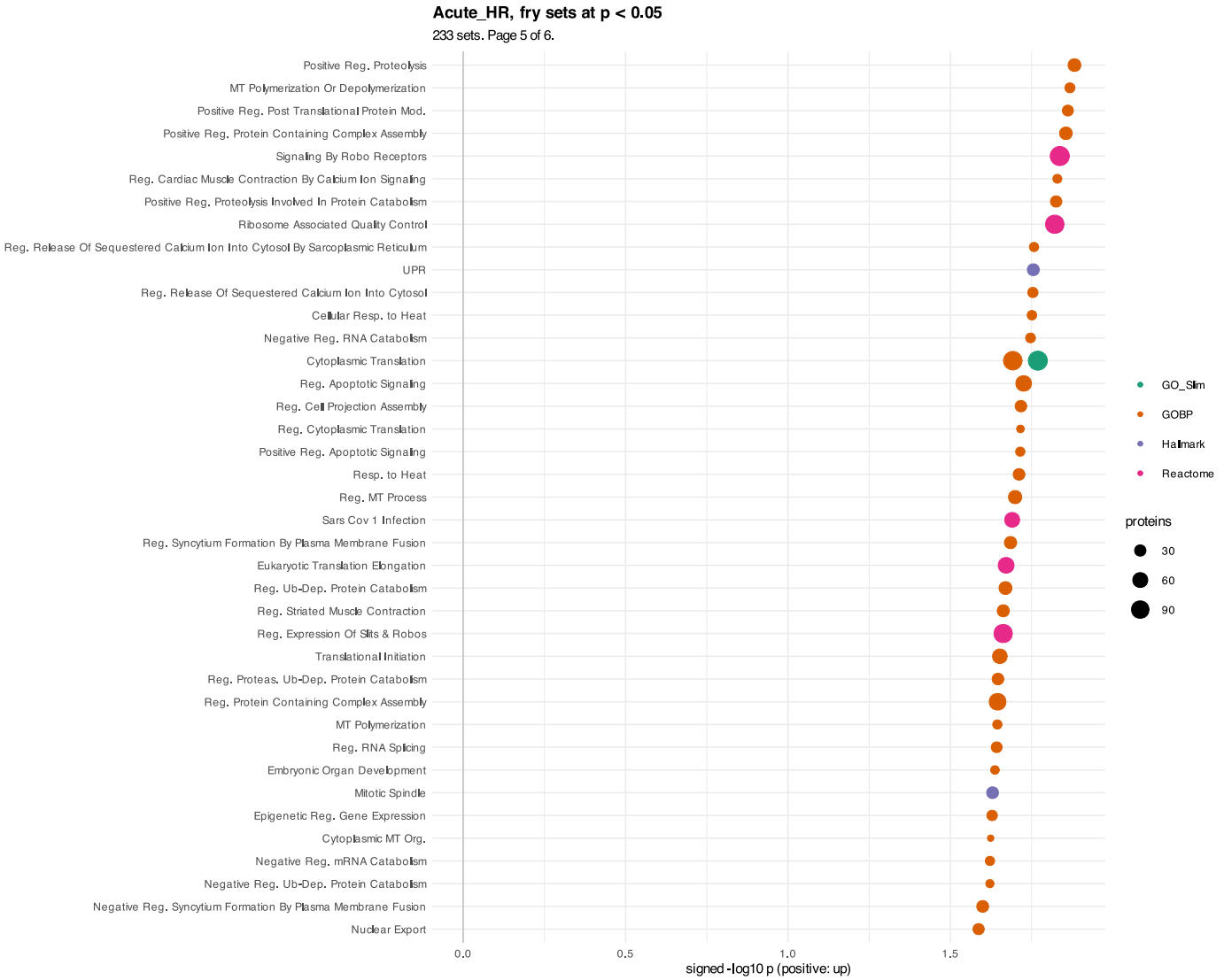


Figure 21: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

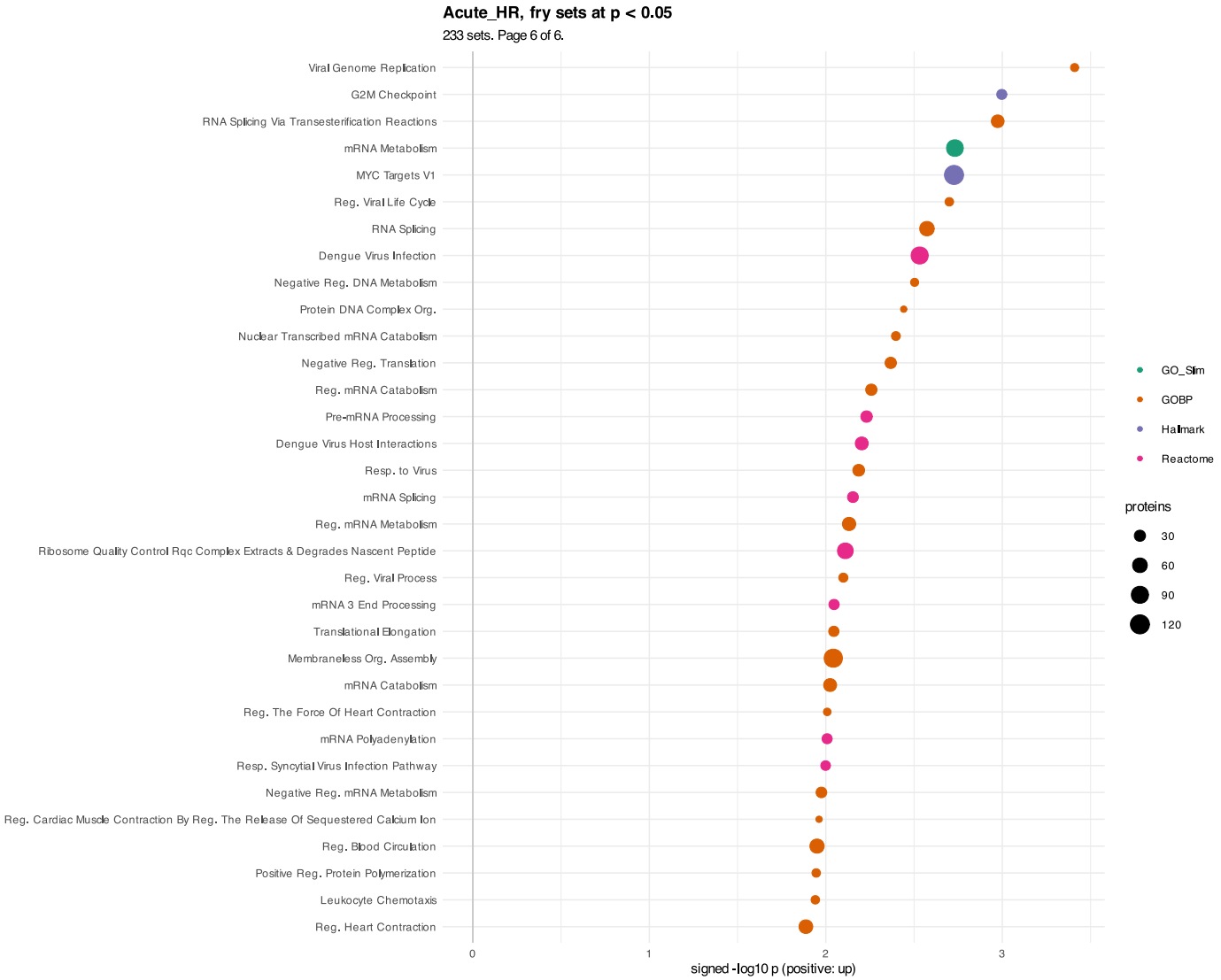


Figure 22: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

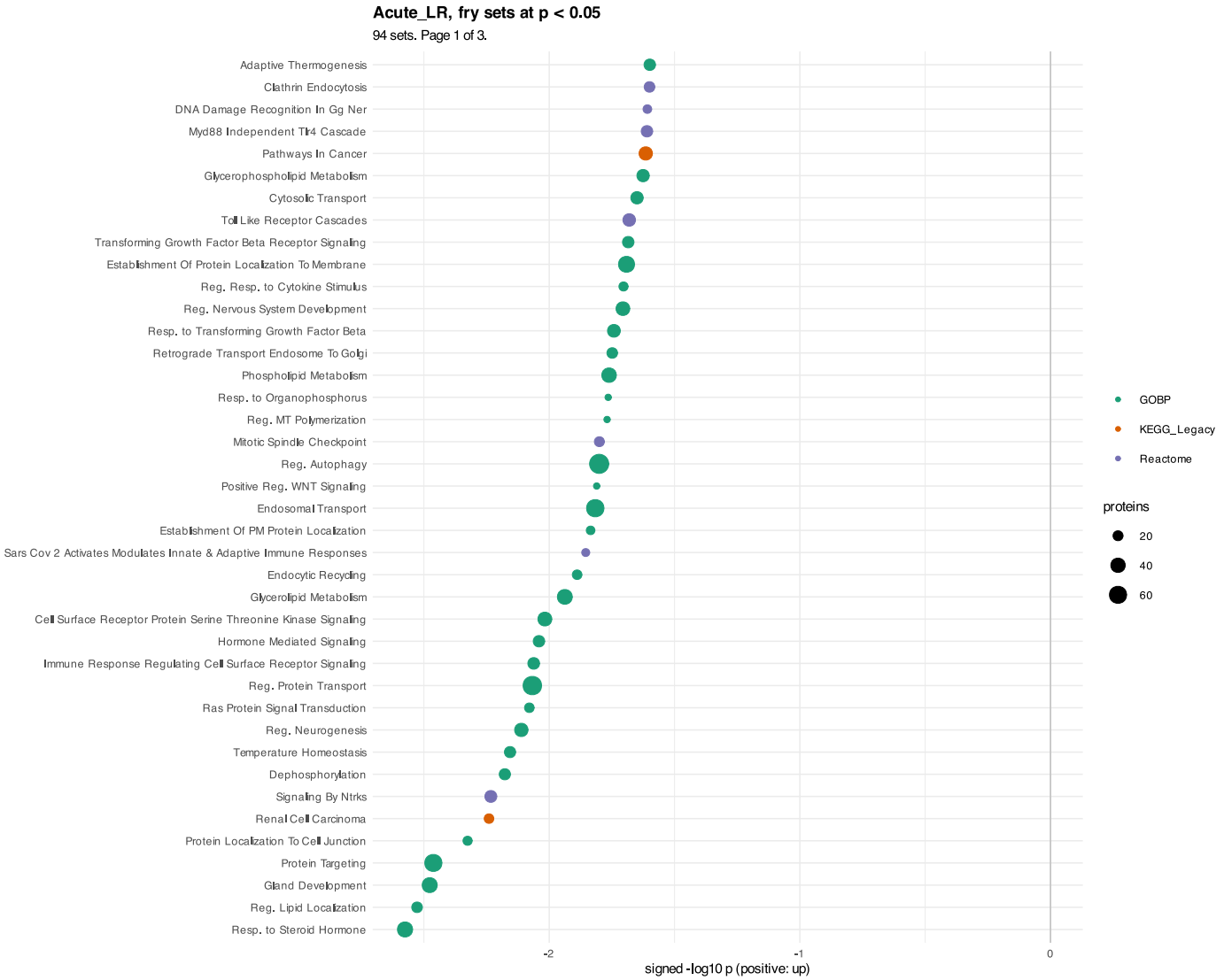


Figure 23: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

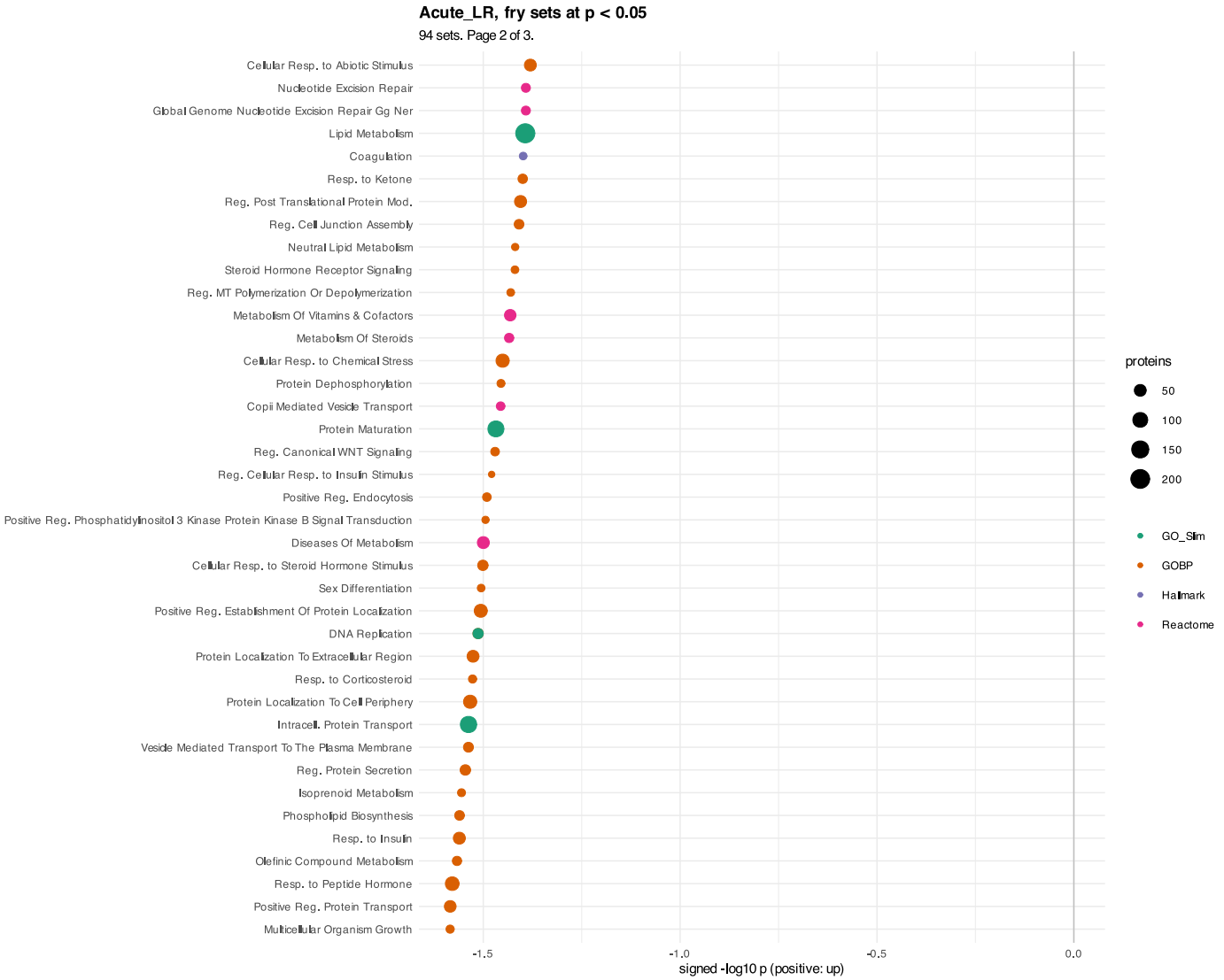


Figure 24: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

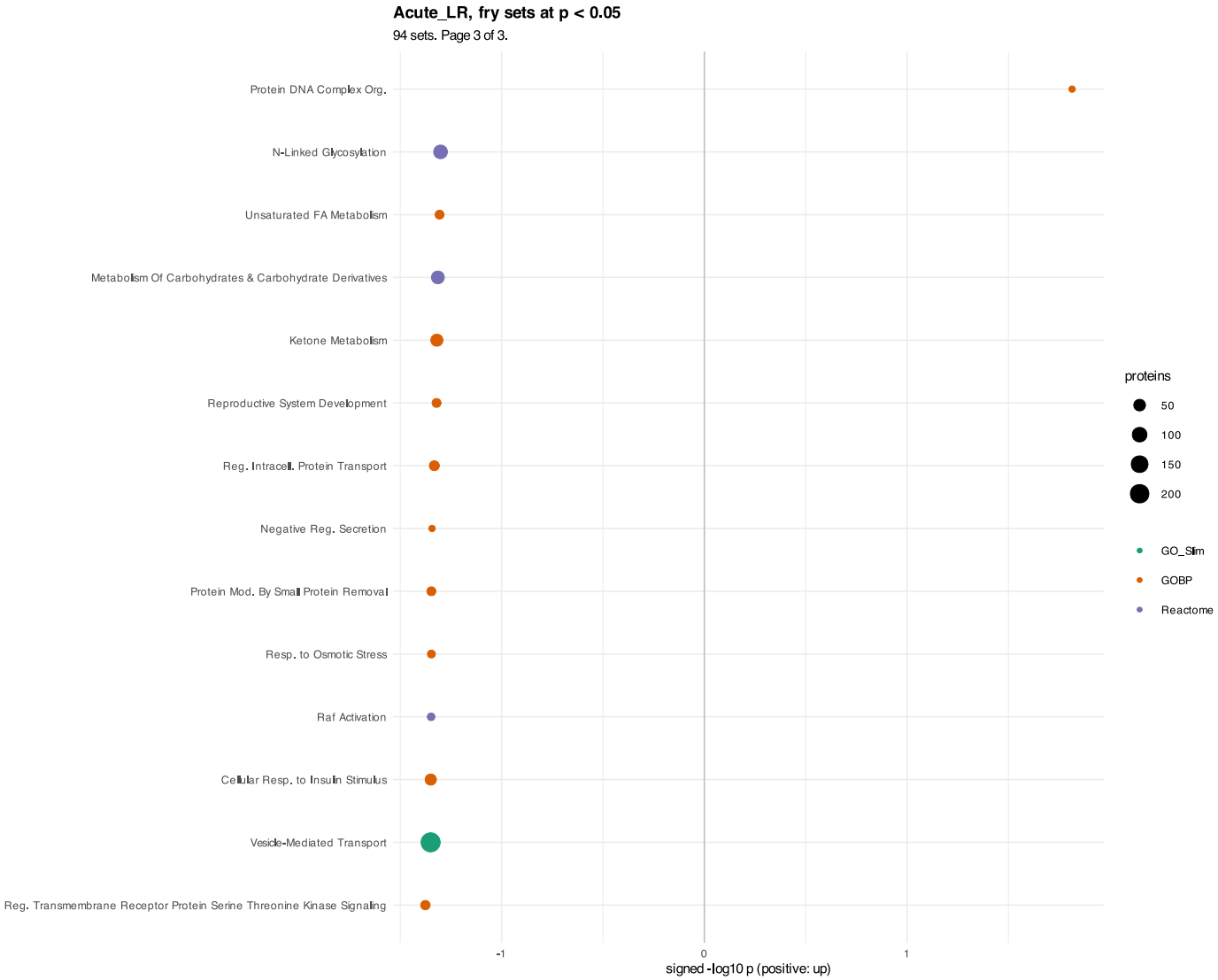


Figure 25: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

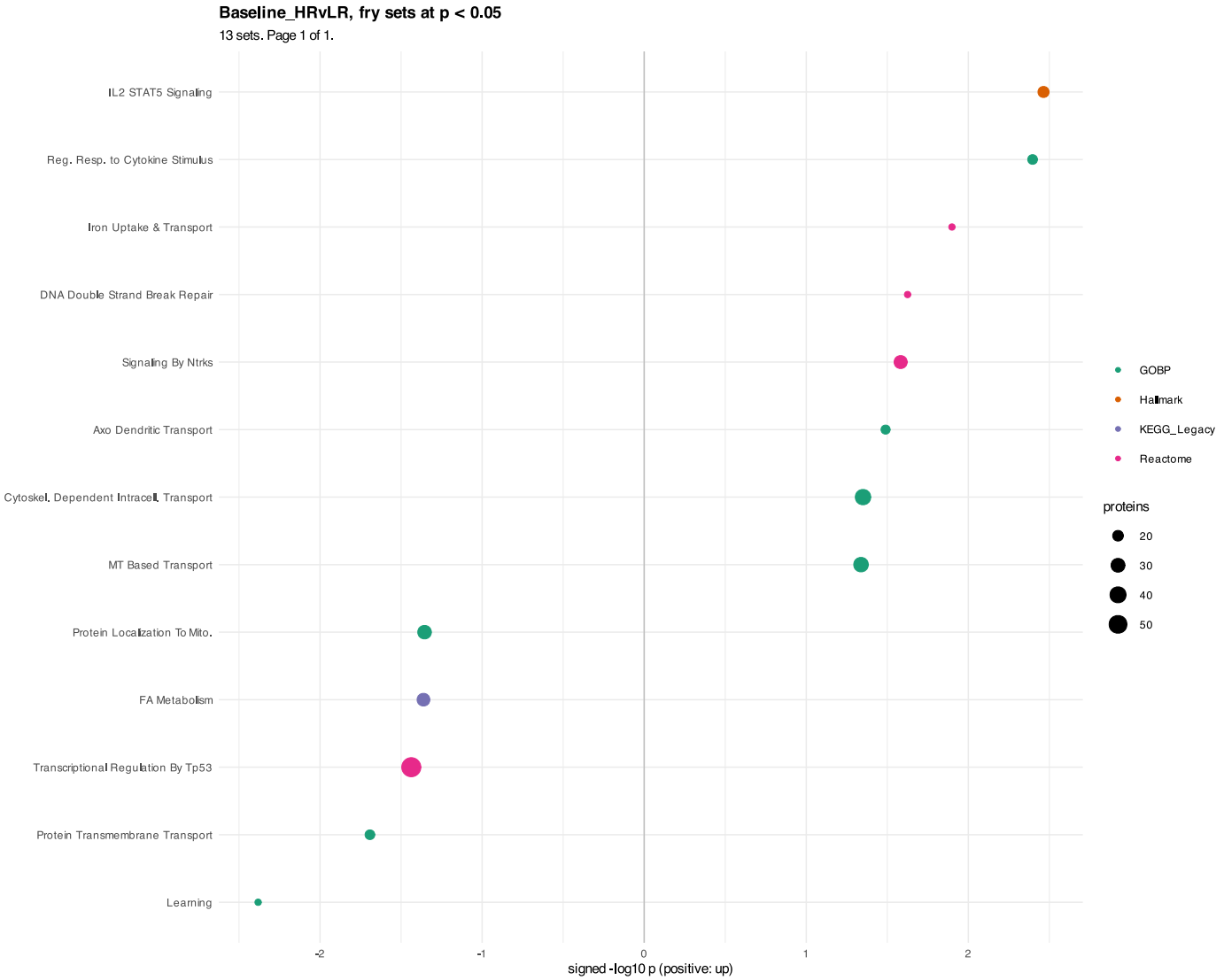


Figure 26: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

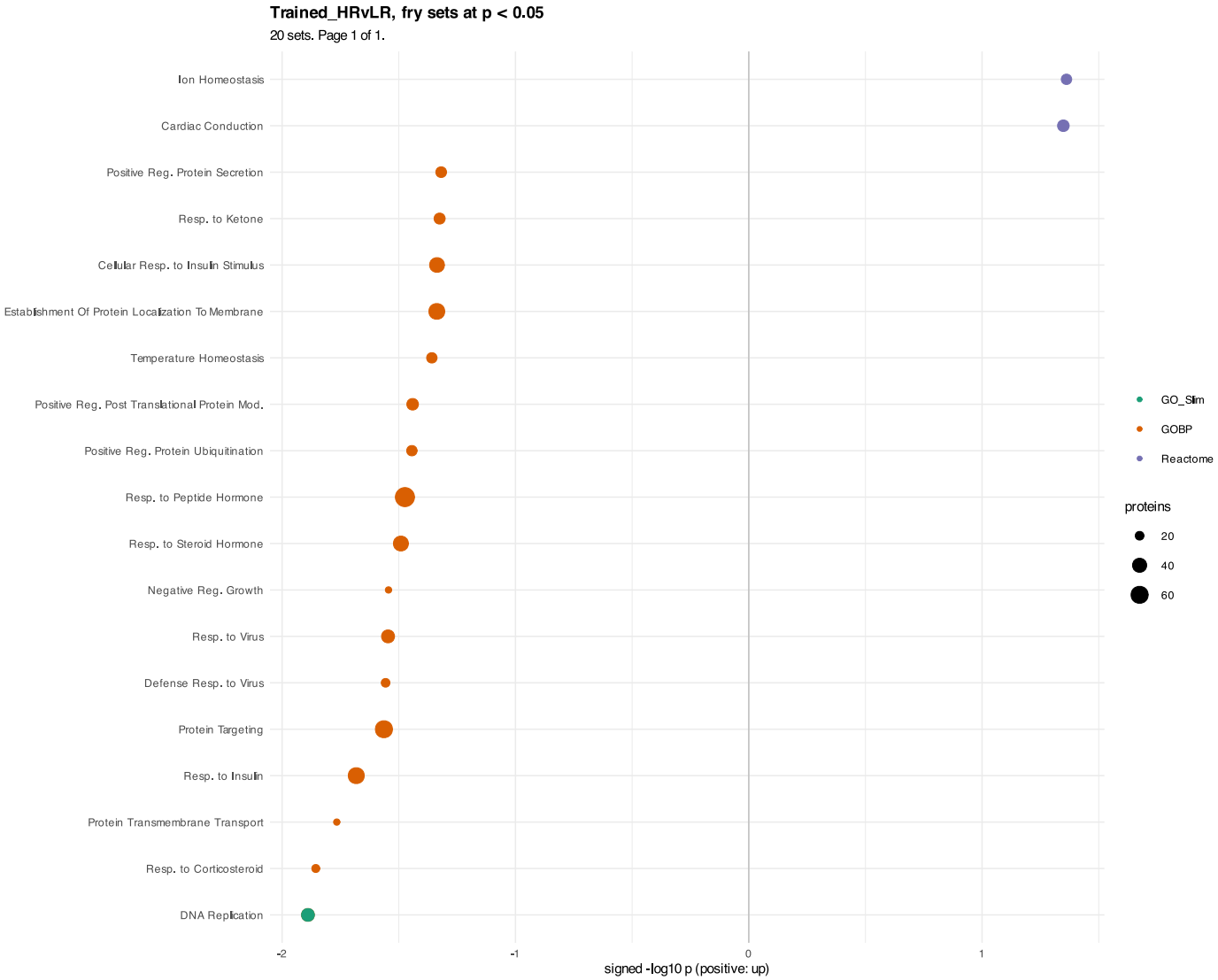


Figure 27: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

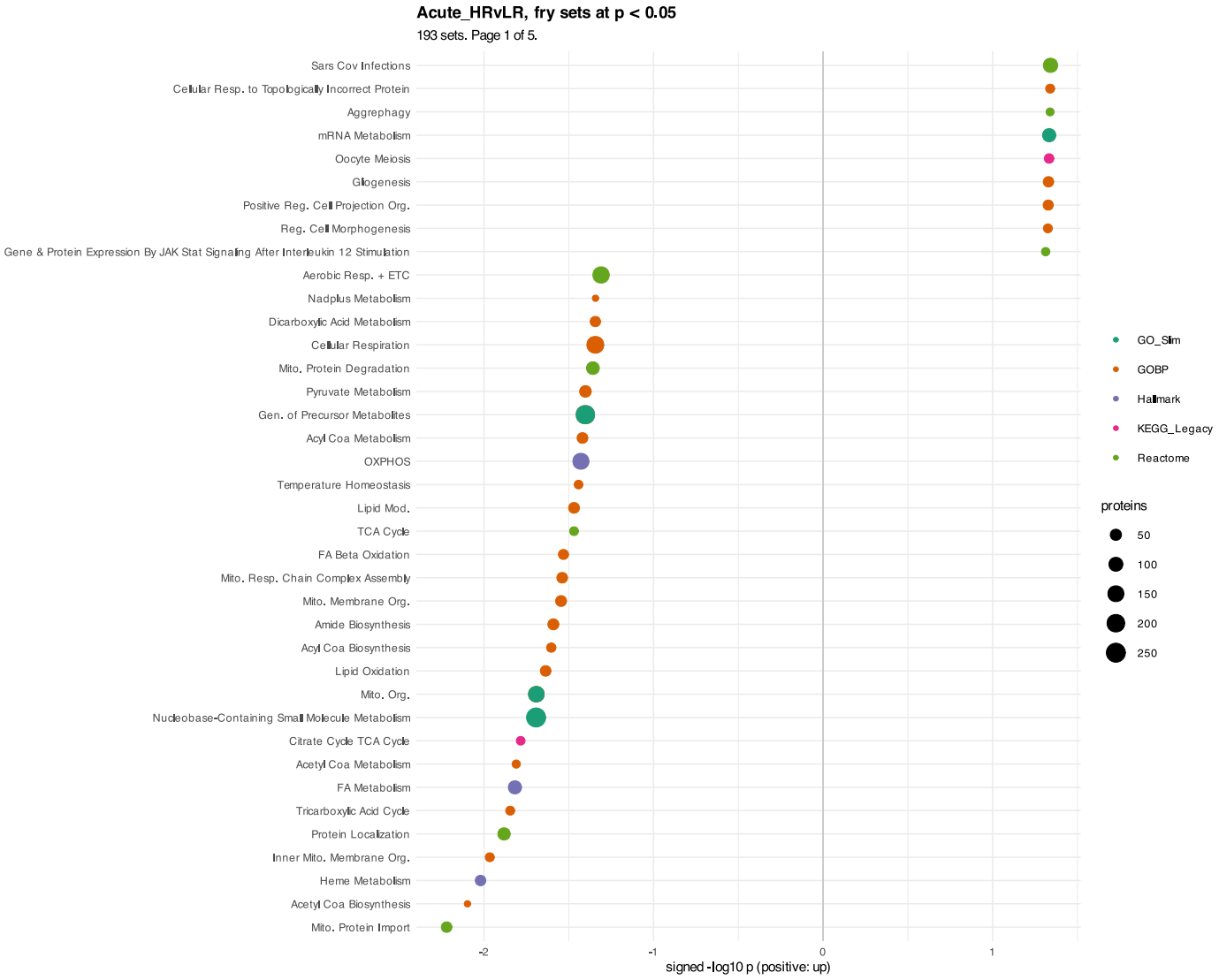


Figure 28: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

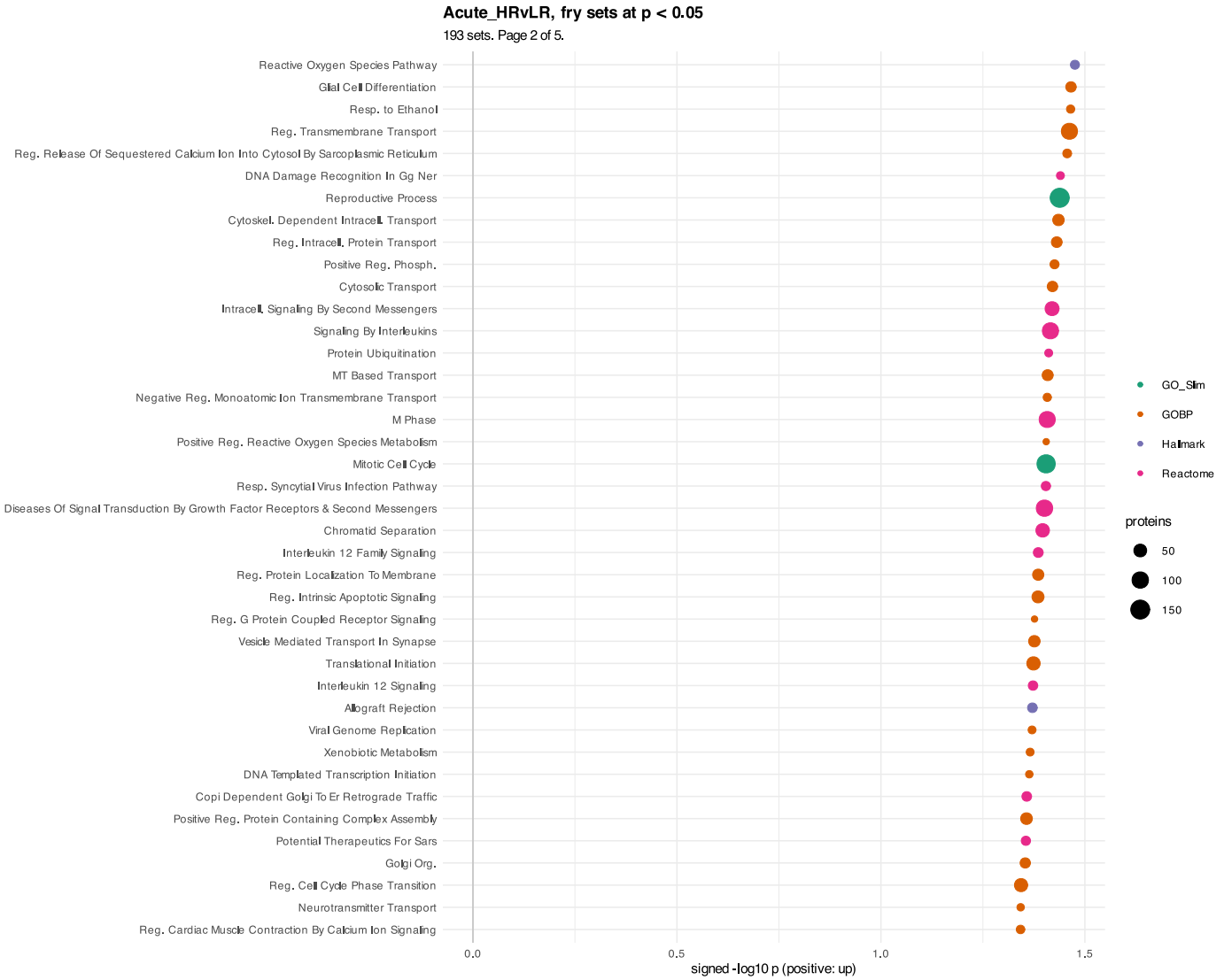


Figure 29: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

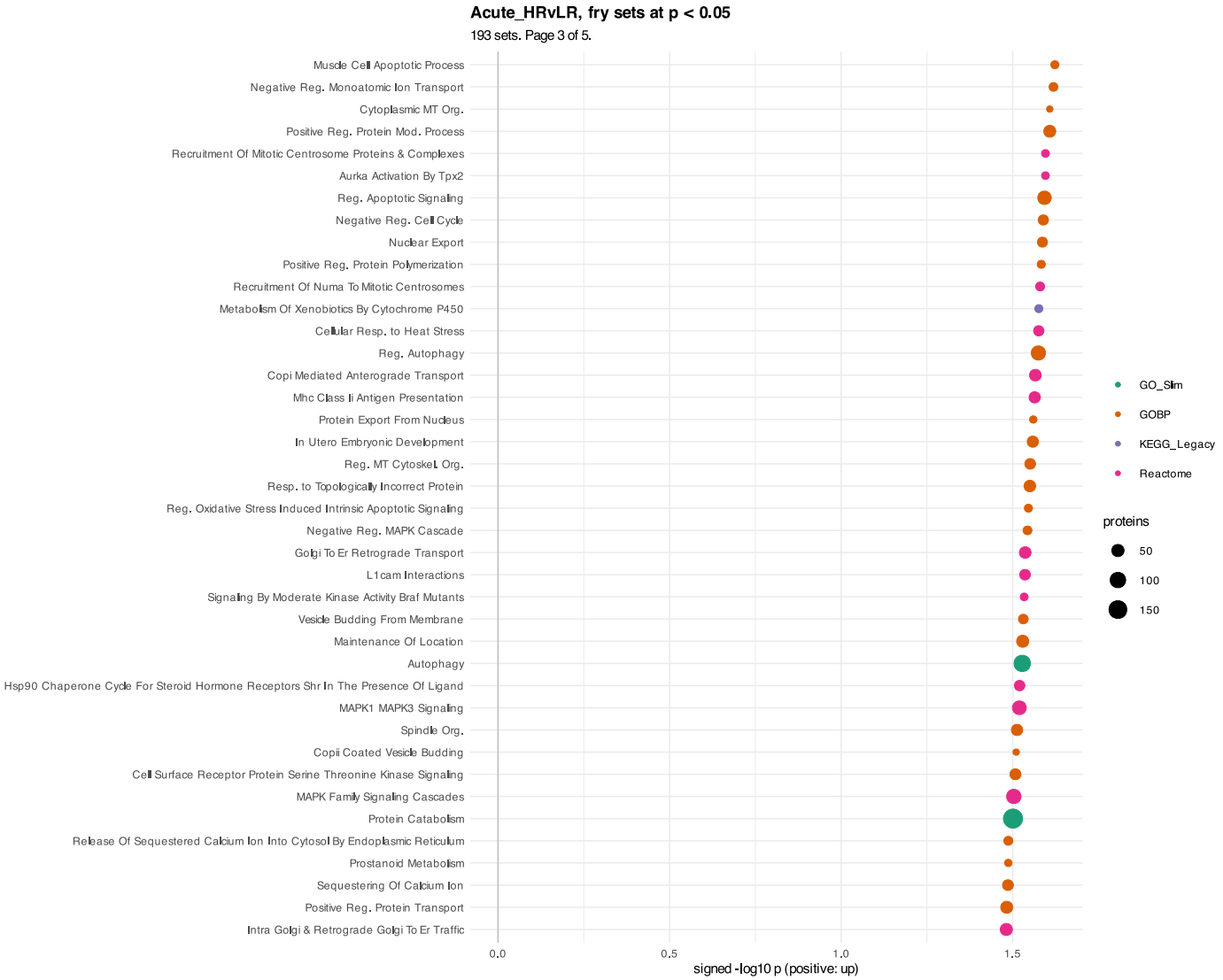


Figure 30: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

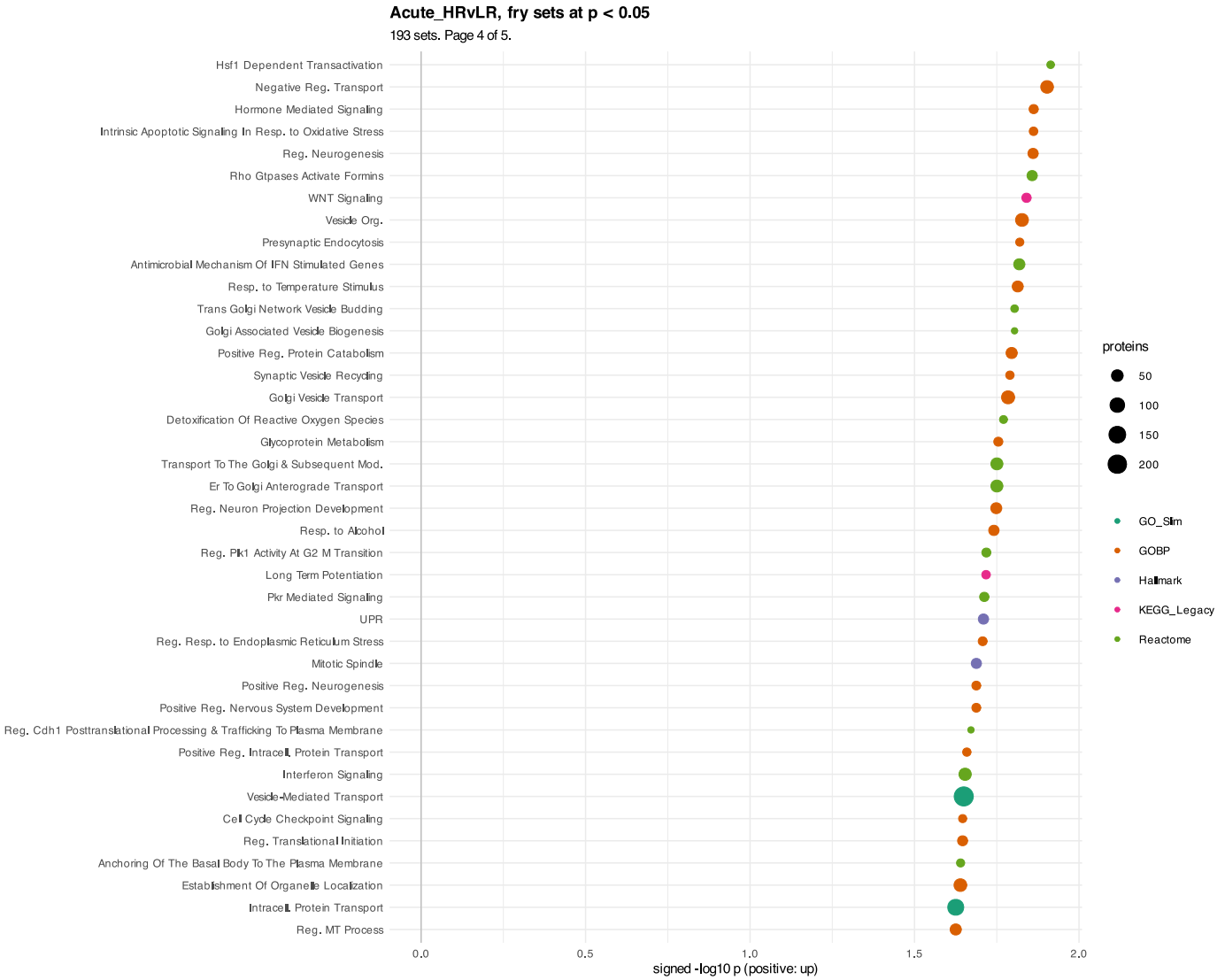


Figure 31: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

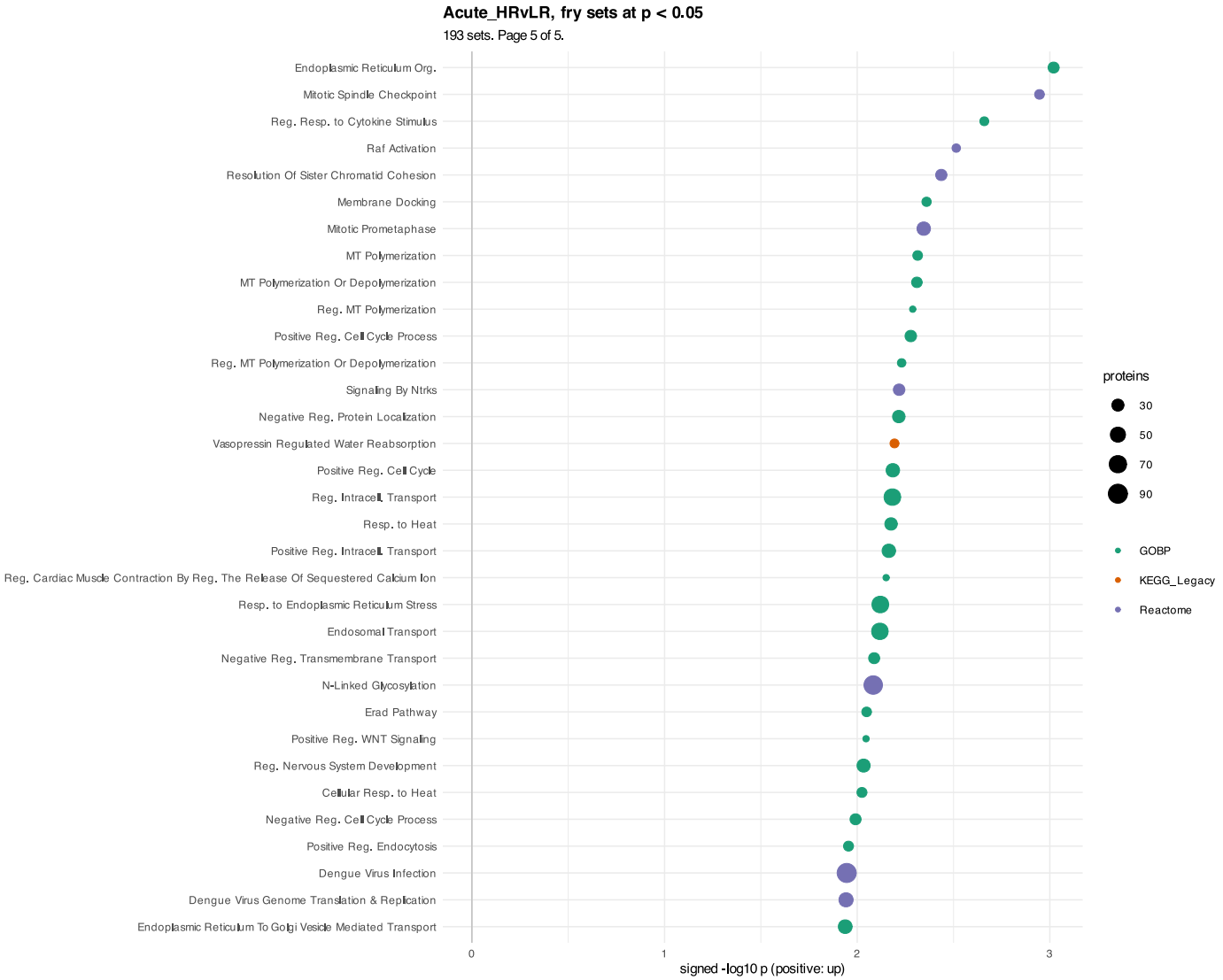


Figure 32: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

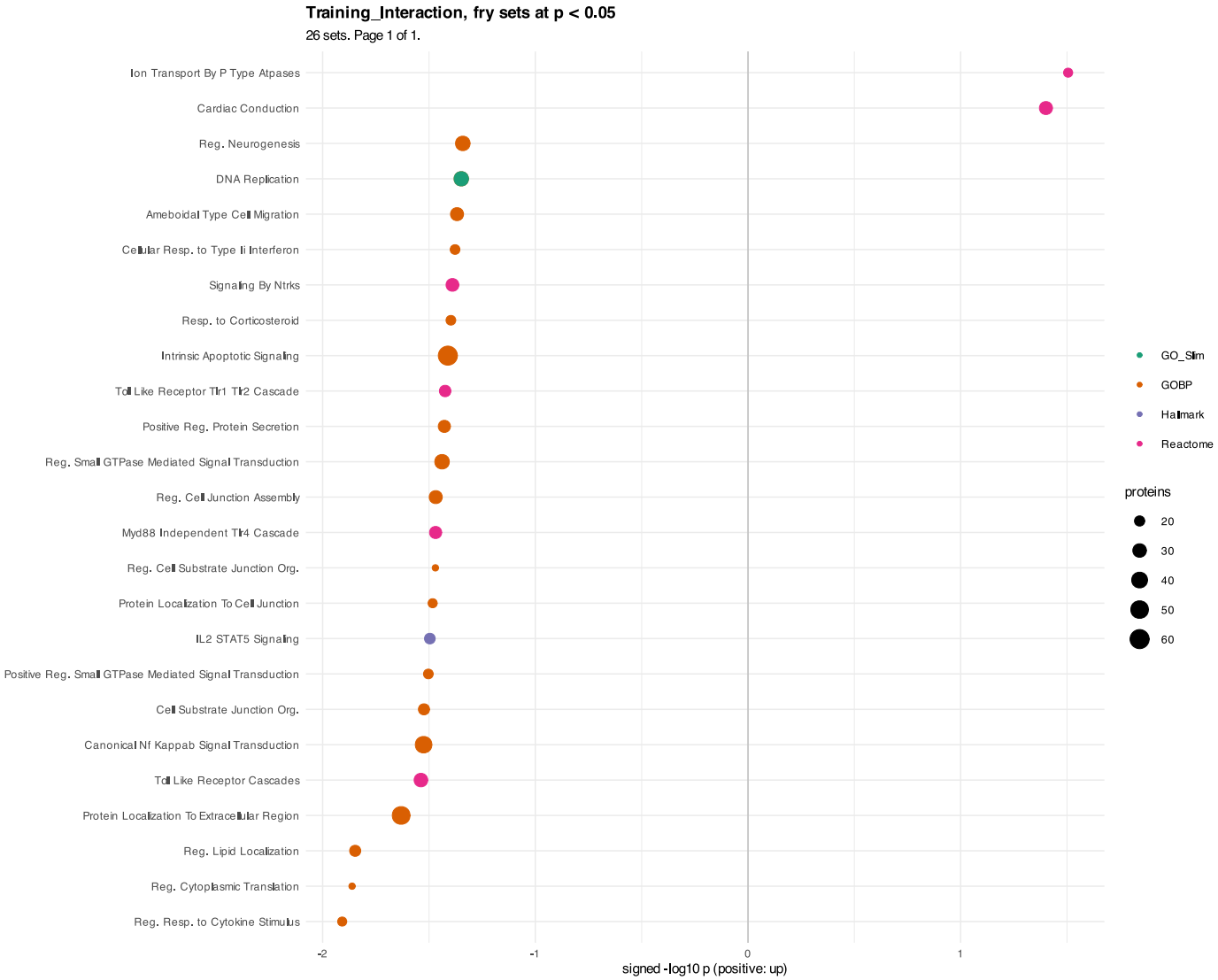


Figure 33: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

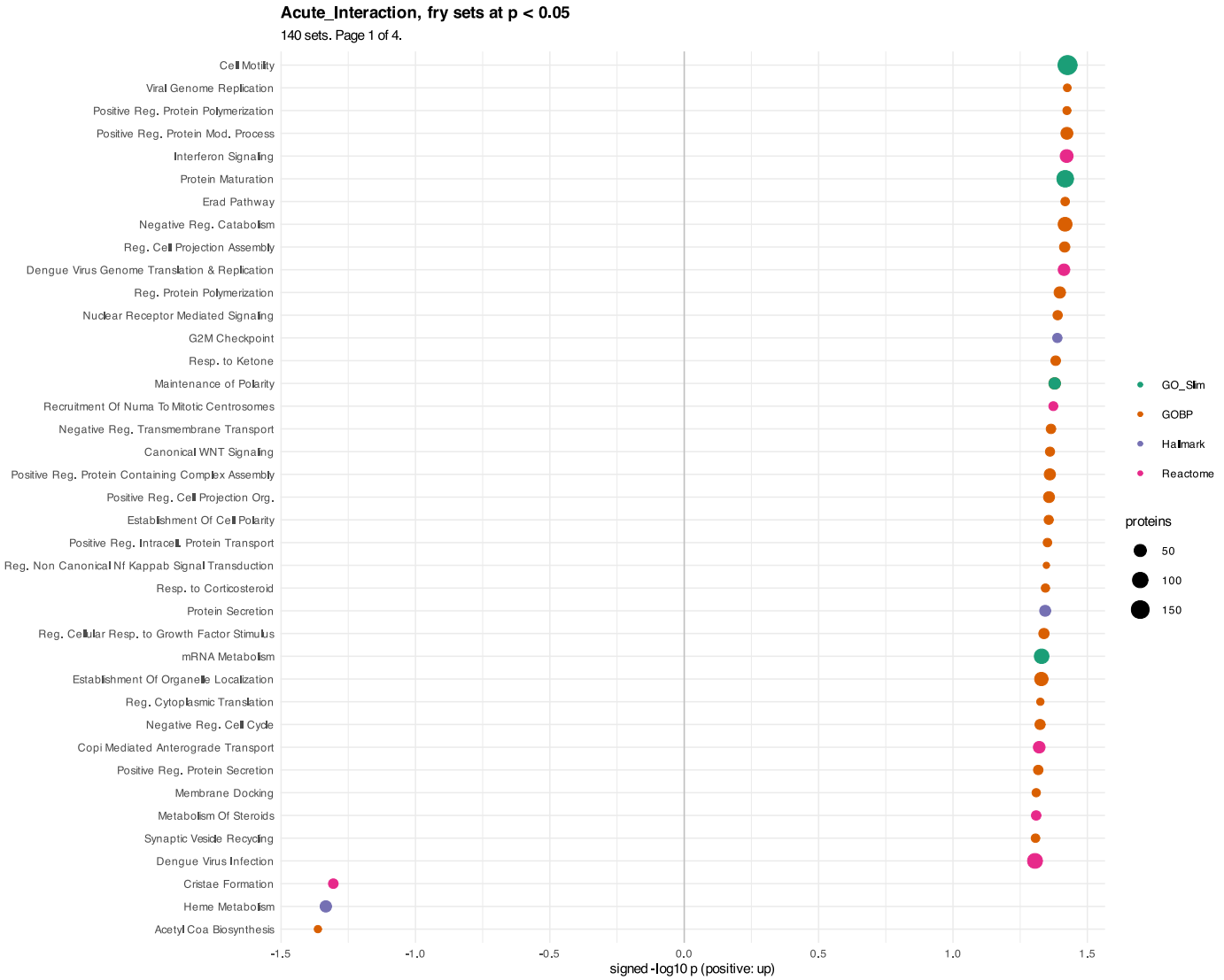


Figure 34: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

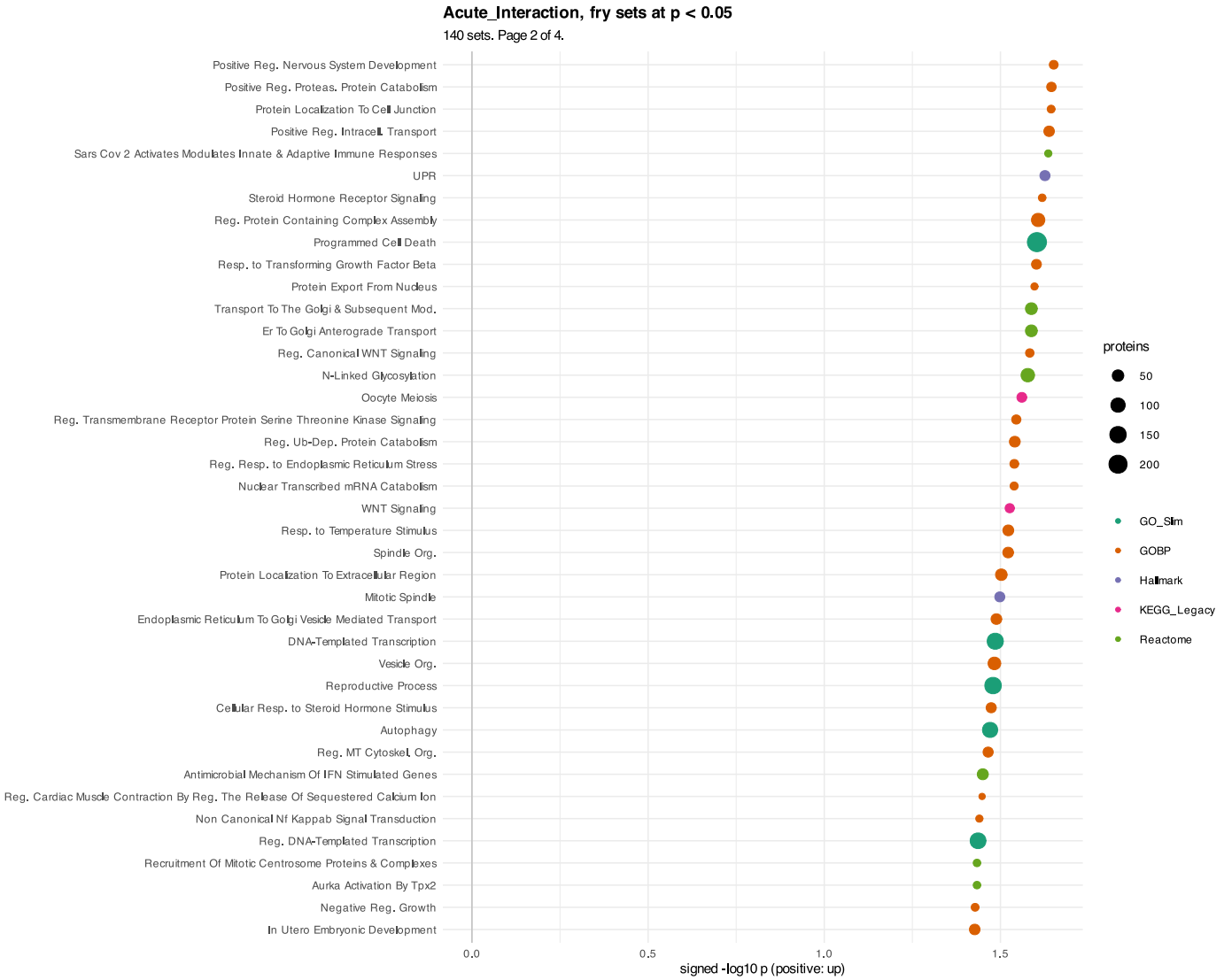


Figure 35: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

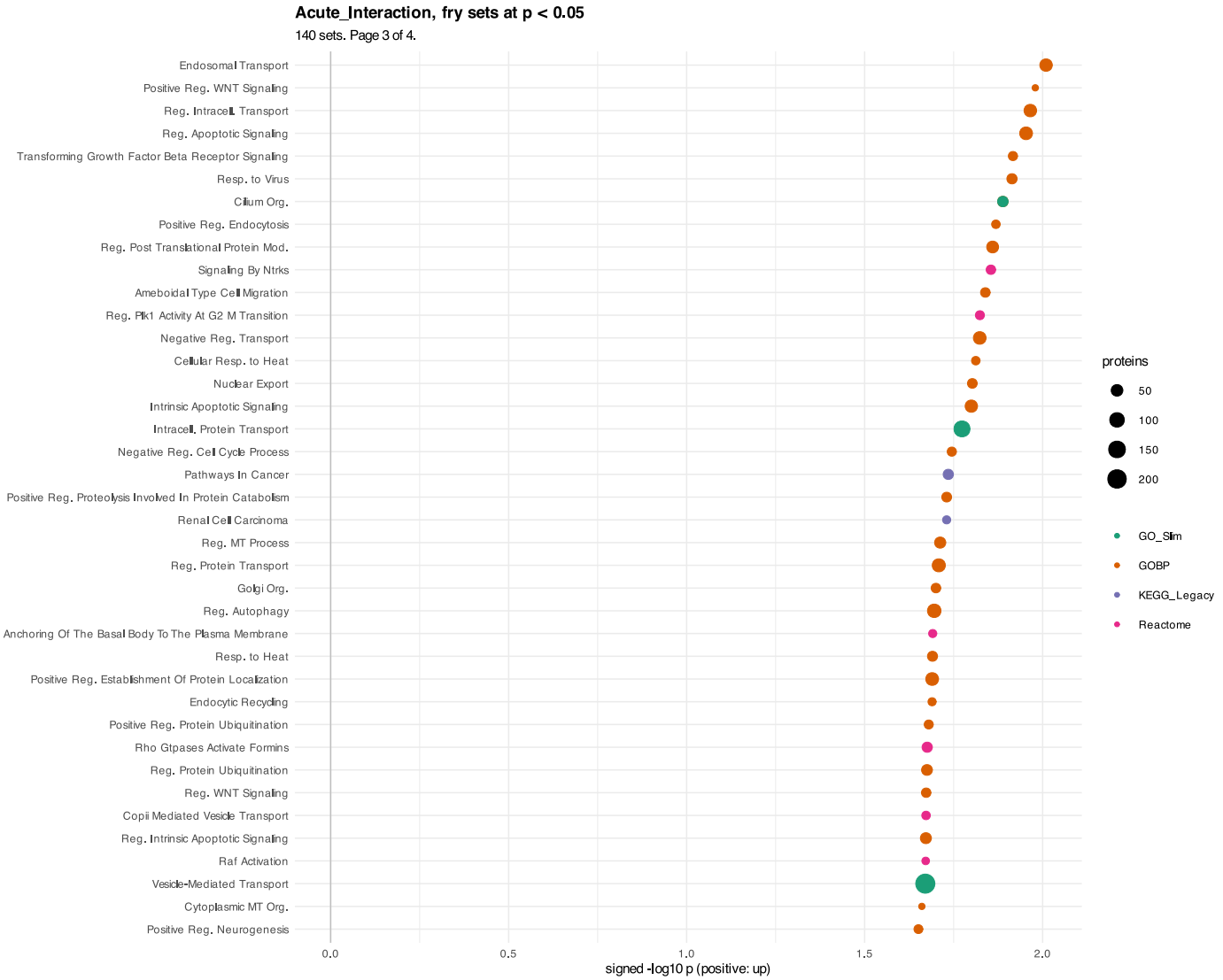


Figure 36: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

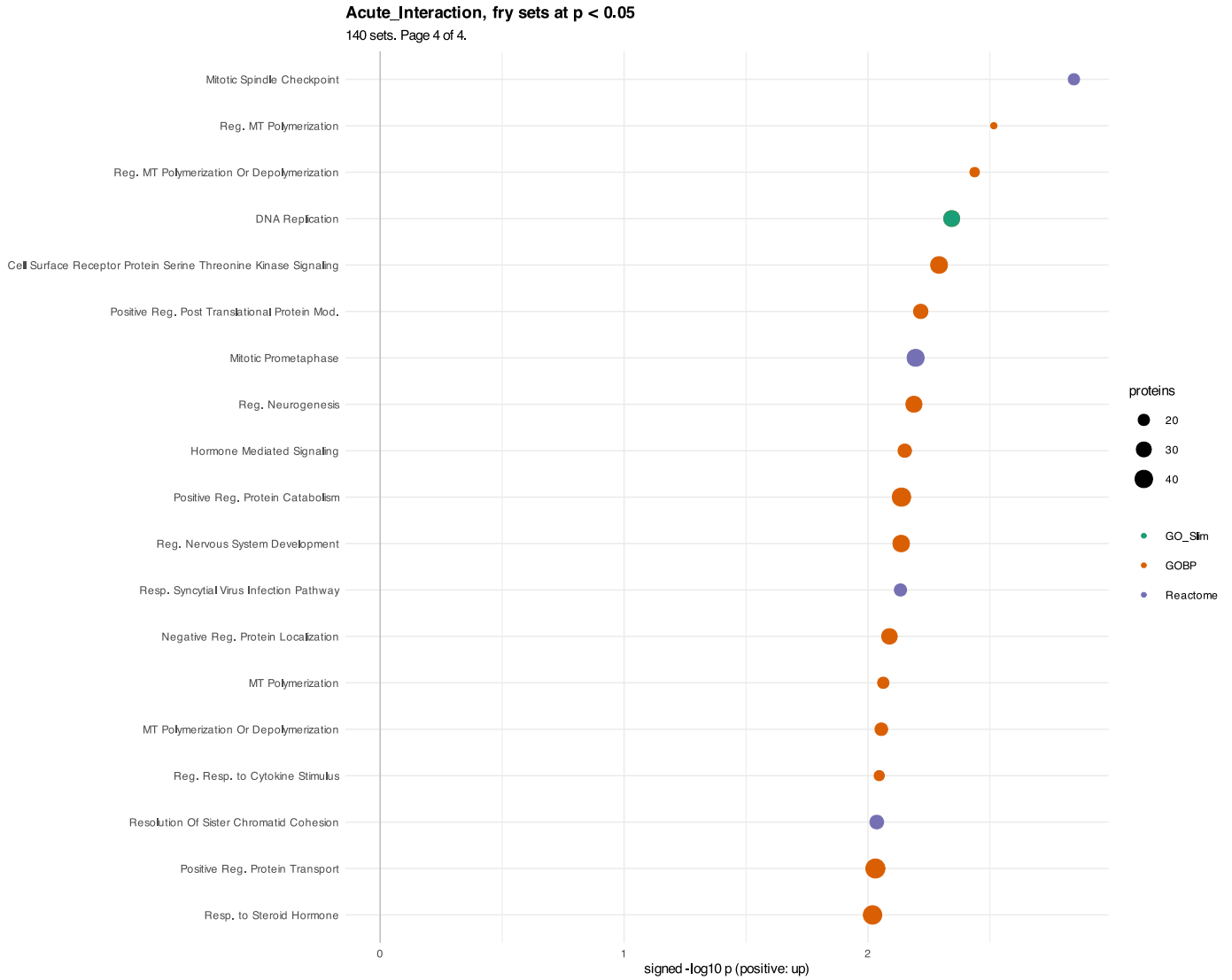


Figure 37: Sets called per contrast: fgsea collapse survivors, then fry sets at p < 0.05.

Every set at a glance

As for proteins: every set's singscore across biopsies (centred on the set's mean), then its fgsea NES across the nine contrasts, one pair of pages per collection. Rows follow average-linkage clustering on 1 – Pearson r of singscore within collection, for display only.

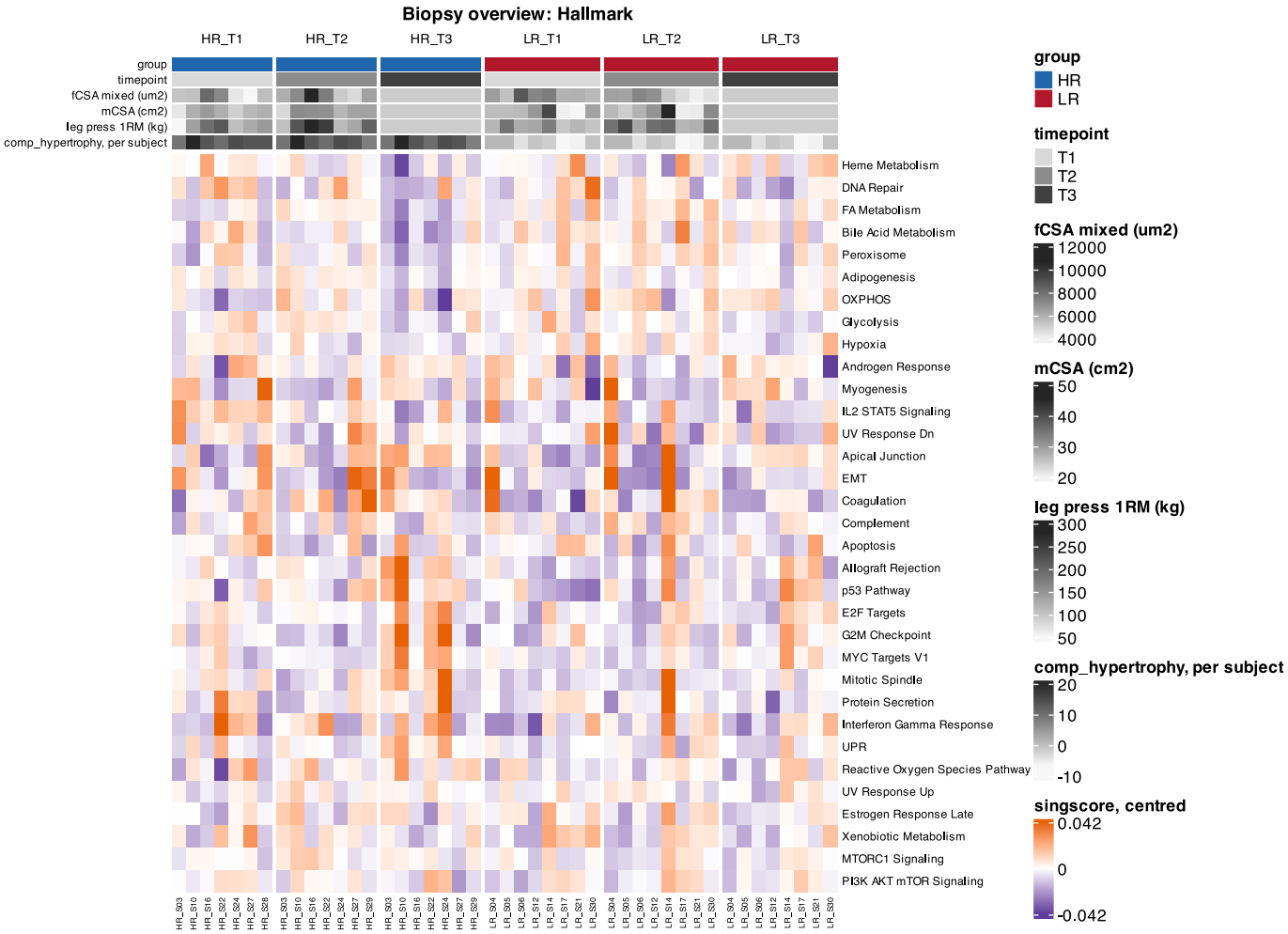
```
catalog <- filter(gs$set_catalog, qualifies)
sample_values <- subset_for_design(readRDS(paths[["singscore"]]), d)$scores[catalog$set_id, ]
feature_label <- with(catalog, set_names(enrichVolcano::ev_clean_label(pathway), set_id))
contrast_results <- set_tests |>
  filter(method == "fgsea") |>
  transmute(feature = set_id, contrast, effect = nes, p, fdr)
row_groups <- split(catalog$set_id, factor(catalog$collection, unique(catalog$collection))) |>
  map(\(ids) ids[hclust(as.dist(1 - cor(t(sample_values[ids, ]))), "average")$order])
report_group_colours <- set_names(c("#2166AC", "#B2182B"), d$groups)
```

```
trait_table <- readr::read_csv(paths[["phenotype"]], show_col_types = FALSE)
biopsy_traits <- trait_table |>
  select(subject, matches("_(pre|post)_")) |>
  pivot_longer(-subject, names_to = c("trait", "when"), names_pattern = "(.*)_(pre|post)_.*") |
>
  pivot_wider(names_from = trait) |>
  transmute(
    subject,
    timepoint = if_else(when == "pre", "T1", "T2"), fcsa_mixed, mcsa, leg_press_1rm
  )
columns <- as_tibble(complete$metadata) |>
  transmute(sample_id, subject, group = as.character(group), timepoint =
as.character(timepoint)) |>
  mutate(group_time = factor(
    paste(group, timepoint, sep = "_"),
    paste(rep(d$groups, each = 3), c("T1", "T2", "T3"), sep = "_")
  )) |>
  left_join(biopsy_traits, by = c("subject", "timepoint")) |>
  left_join(select(trait_table, subject, comp_hypertrophy), by = "subject") |>
  arrange(group_time, subject)
stopifnot(!anyDuplicated(columns$sample_id), setequal(columns$sample_id,
colnames(sample_values)))
lengths(row_groups)
```

```
   Hallmark    Reactome KEGG_Legacy        GOBP     GO_Slim
         33         328          62         897          58
```

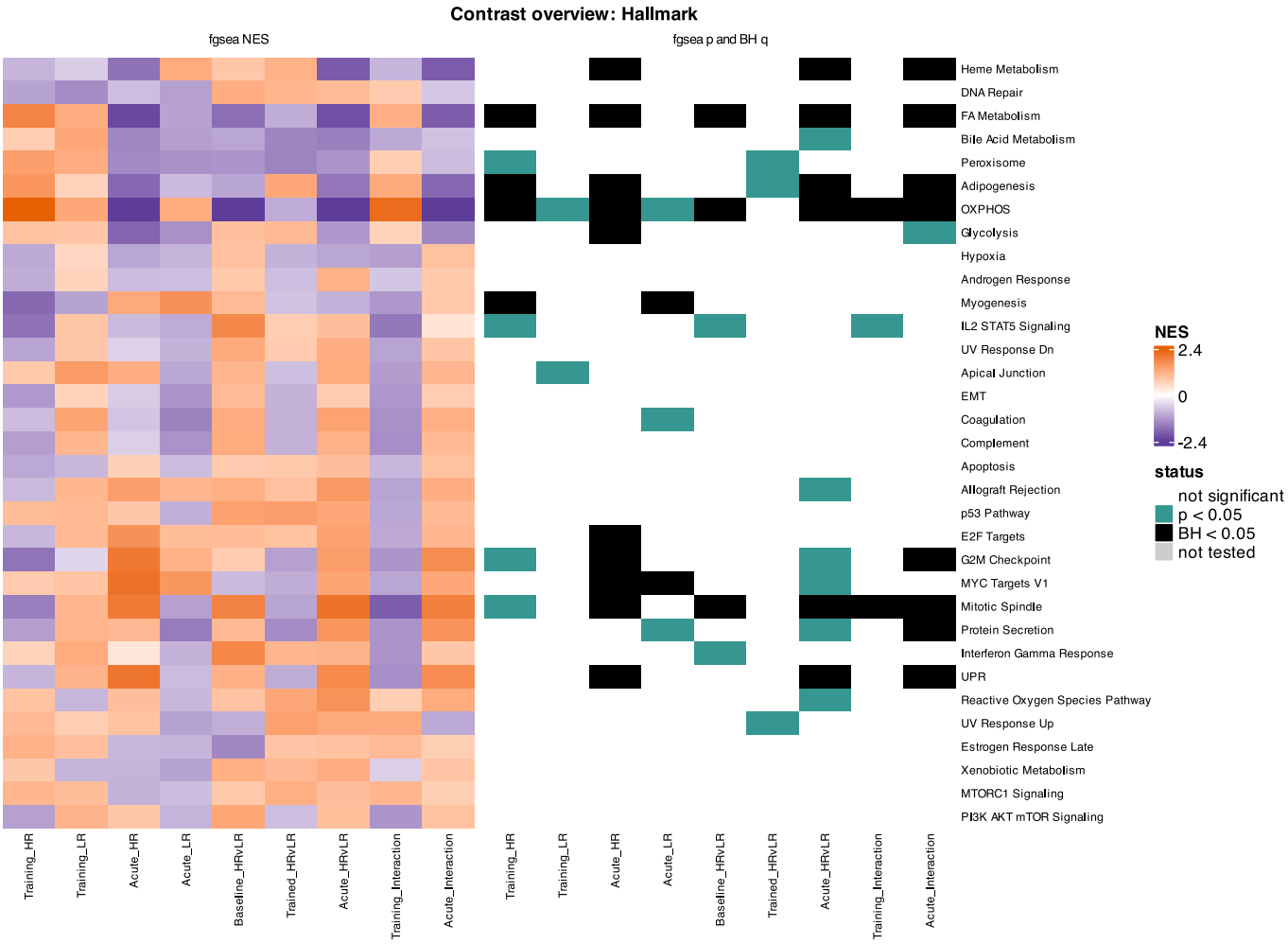
```
grey_ramp <- \(x) circlize::colorRamp2(range(x, na.rm = TRUE), c("#F7F7F7", "#252525"))
tracks <- list(
  "fCSA mixed (um2)" = columns$fcsa_mixed,
  "mCSA (cm2)" = columns$mcsa, "leg press 1RM (kg)" = columns$leg_press_1rm,
  "comp_hypertrophy, per subject" = columns$comp_hypertrophy
)
sample_annotation <- HeatmapAnnotation(
  df = data.frame(
    group = columns$group, timepoint = columns$timepoint, tracks, check.names = FALSE
  ),
  col = c(
    list(group = report_group_colours, timepoint = c(T1 = "grey85", T2 = "grey55", T3 =
"grey25")),
    map(tracks, grey_ramp)
  ),
  na_col = "grey80", annotation_name_gp = gpar(fontsize = 7), annotation_name_side = "left",
  simple_anno_size = unit(3, "mm")
)
effect_limit <- cap(contrast_results$effect)
overviews <- imap(row_groups, \(rows, group) {
  m <- sample_values[rows, columns$sample_id, drop = FALSE]
  m <- sweep(m, 1, rowMeans(m))
```

```
  limit <- cap(m)
  labelled_heat <- \(...) make_report_heatmap(..., labels = feature_label)
  biopsy_map <- labelled_heat(m, "biopsies", c(-limit, 0, limit), "singscore, centred",
    row_names = TRUE, column_title = character(0), top_annotation = sample_annotation,
    column_split = columns$group_time, cluster_column_slices = FALSE,
    column_labels = columns$subject, column_names_gp = gpar(fontsize = 5)
  )
  contrast_grid <- \(value) {
    results_to_matrix(contrast_results, rows, contrast_names, "contrast", value)
  }
  contrast_maps <- labelled_heat(
    contrast_grid(\(d) d$effect), "fgsea NES", c(-effect_limit, 0, effect_limit), "NES"
  ) +
    labelled_heat(contrast_grid(\(d) status_of(d$p, d$fdr)), "fgsea p and BH q", row_names =
TRUE)
  list(
    heatmap_page(biopsy_map, sprintf("Biopsy overview: %s", group), paste(
      sprintf("All %d sets by %d biopsies.", length(rows), ncol(m)),
      sprintf("Centred singscore; colours stop at %.2g.", limit)
    ), legends = list()),
    heatmap_page(contrast_maps, sprintf("Contrast overview: %s", group), paste(
      "fgsea NES; status: p and BH q within contrast, collections pooled.",
      sprintf("Colours stop at %.2g.", effect_limit)
    ))
  )
}) |>
  list_flatten()
walk(overviews, \(page) page$draw())
```



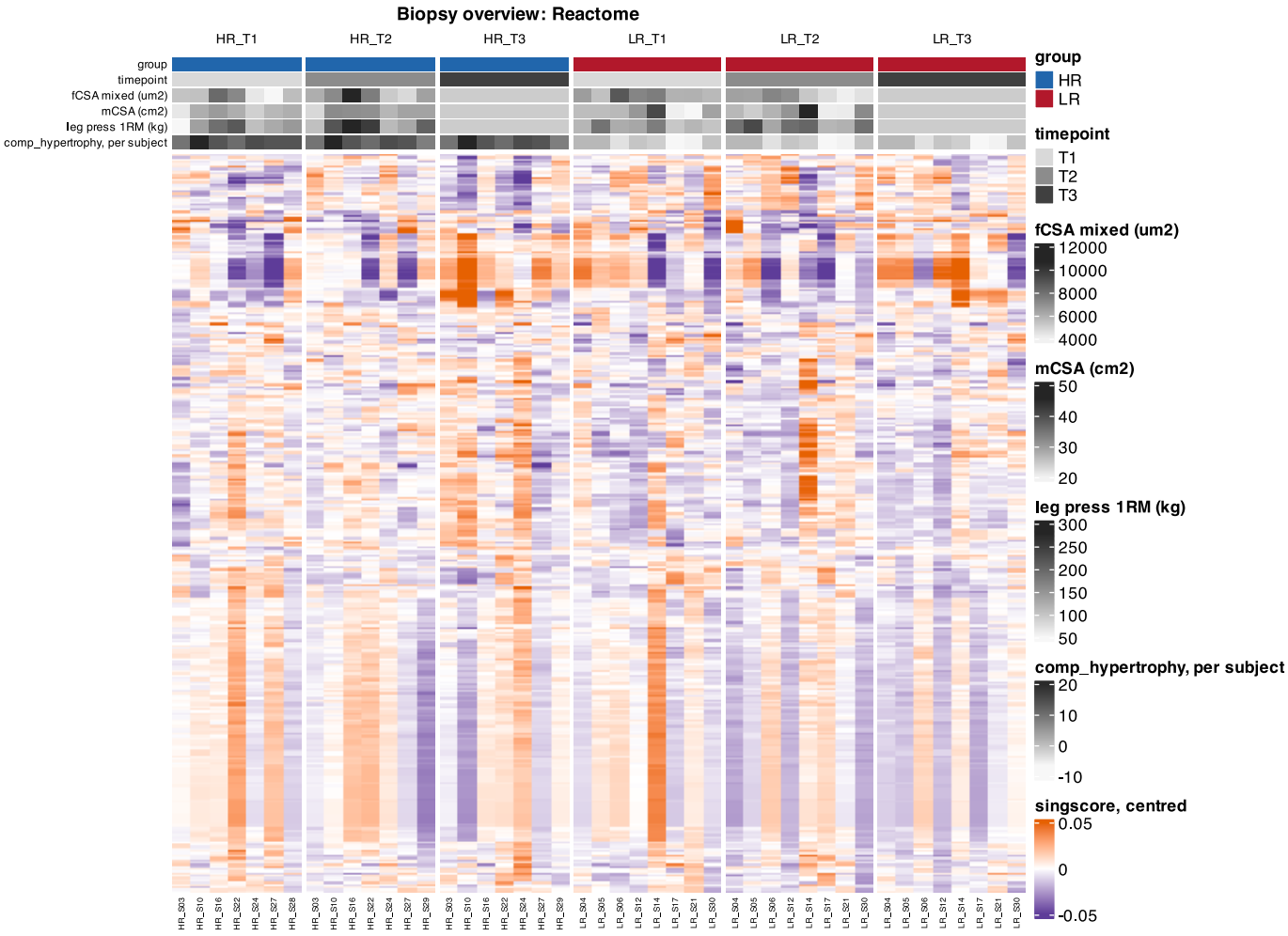
All 33 sets by 45 biopsies. Centred singscore; colours stop at 0.042.

Figure 38: Every set across biopsies, then across the nine contrasts, by collection.



fgsea NES; status: p and BH q within contrast, collections pooled. Colours stop at 2.4.

Figure 39: Every set across biopsies, then across the nine contrasts, by collection.



All 328 sets by 45 biopsies. Centred singscore; colours stop at 0.05.

Figure 40: Every set across biopsies, then across the nine contrasts, by collection.

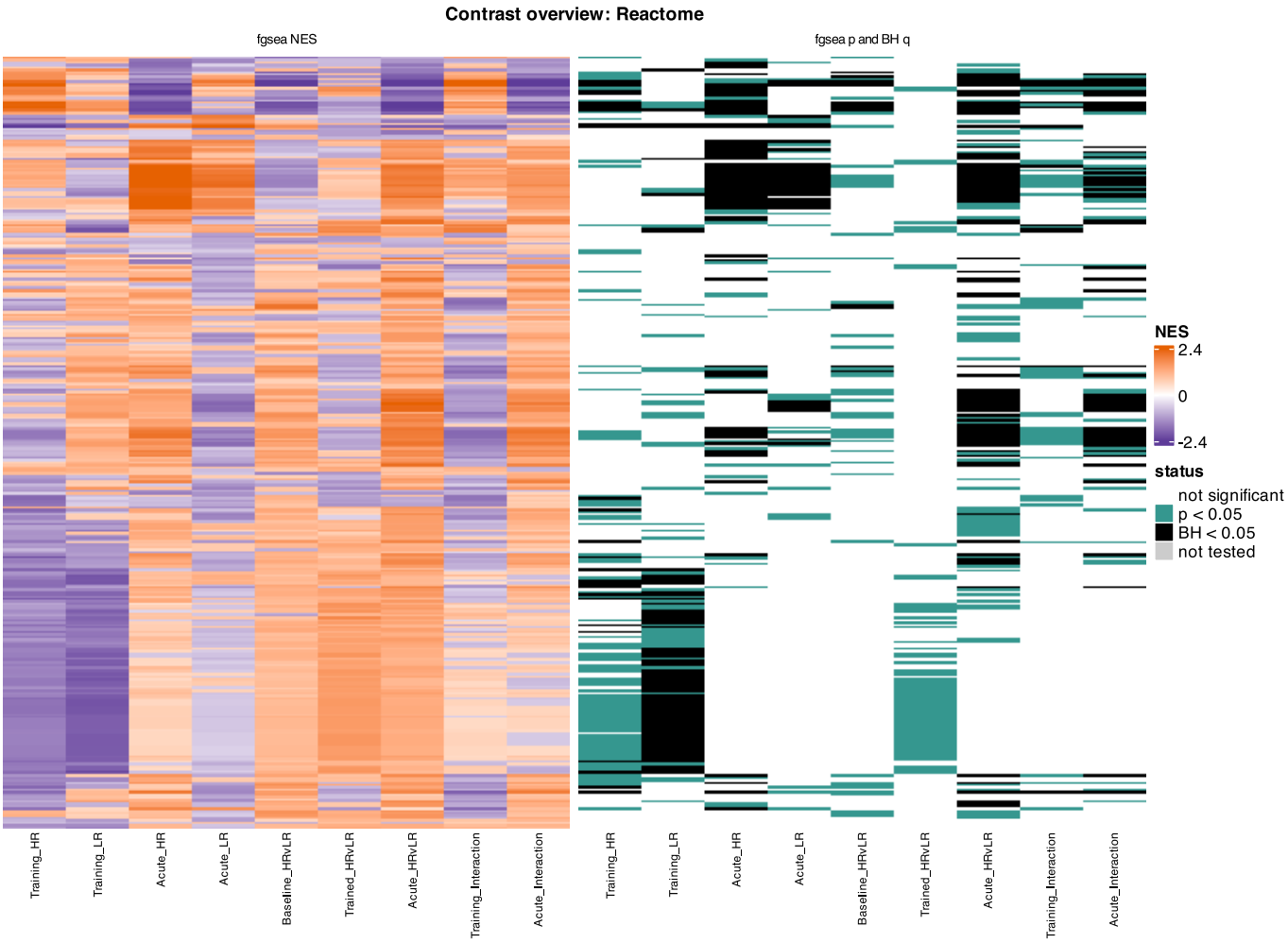
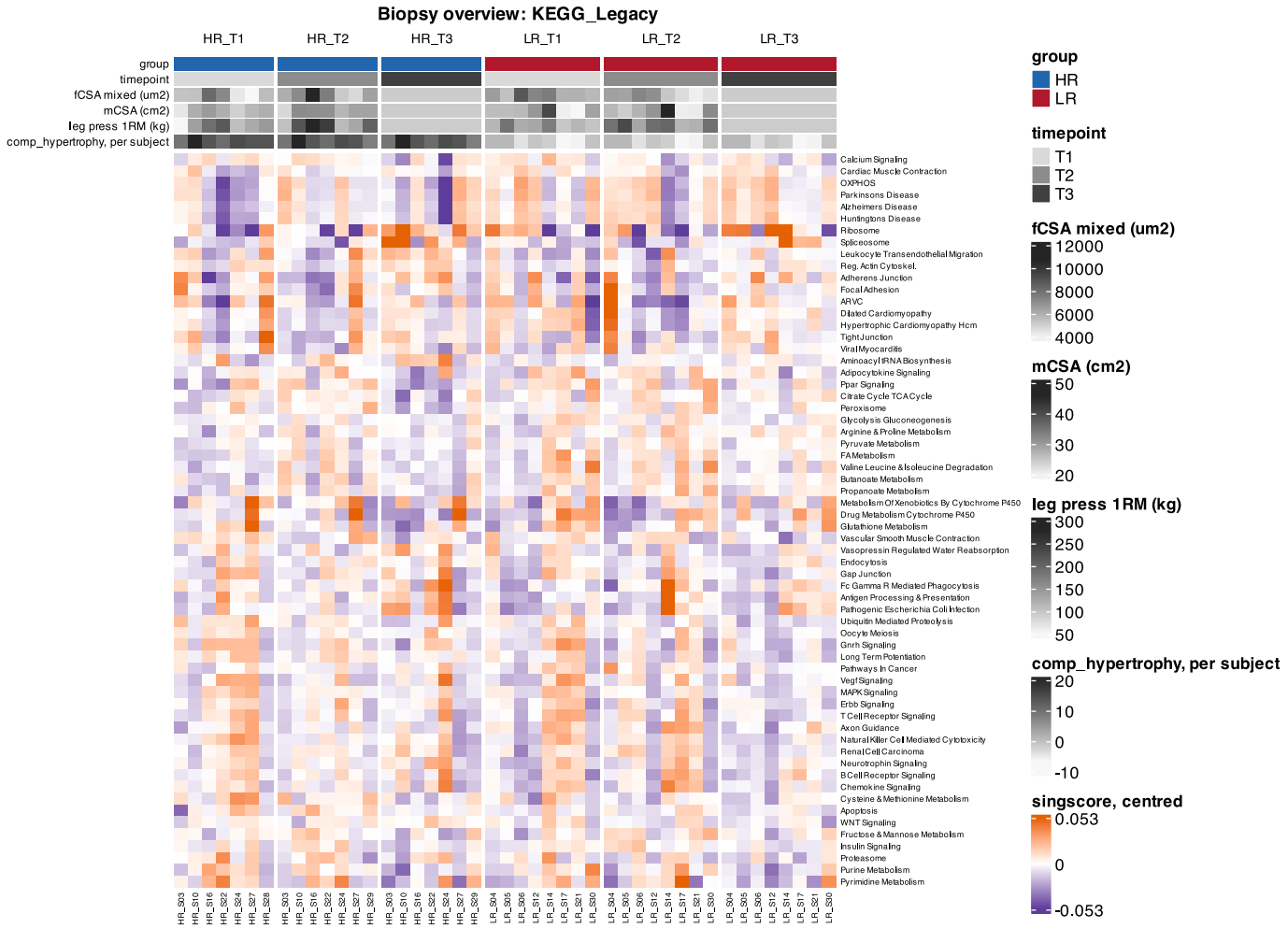
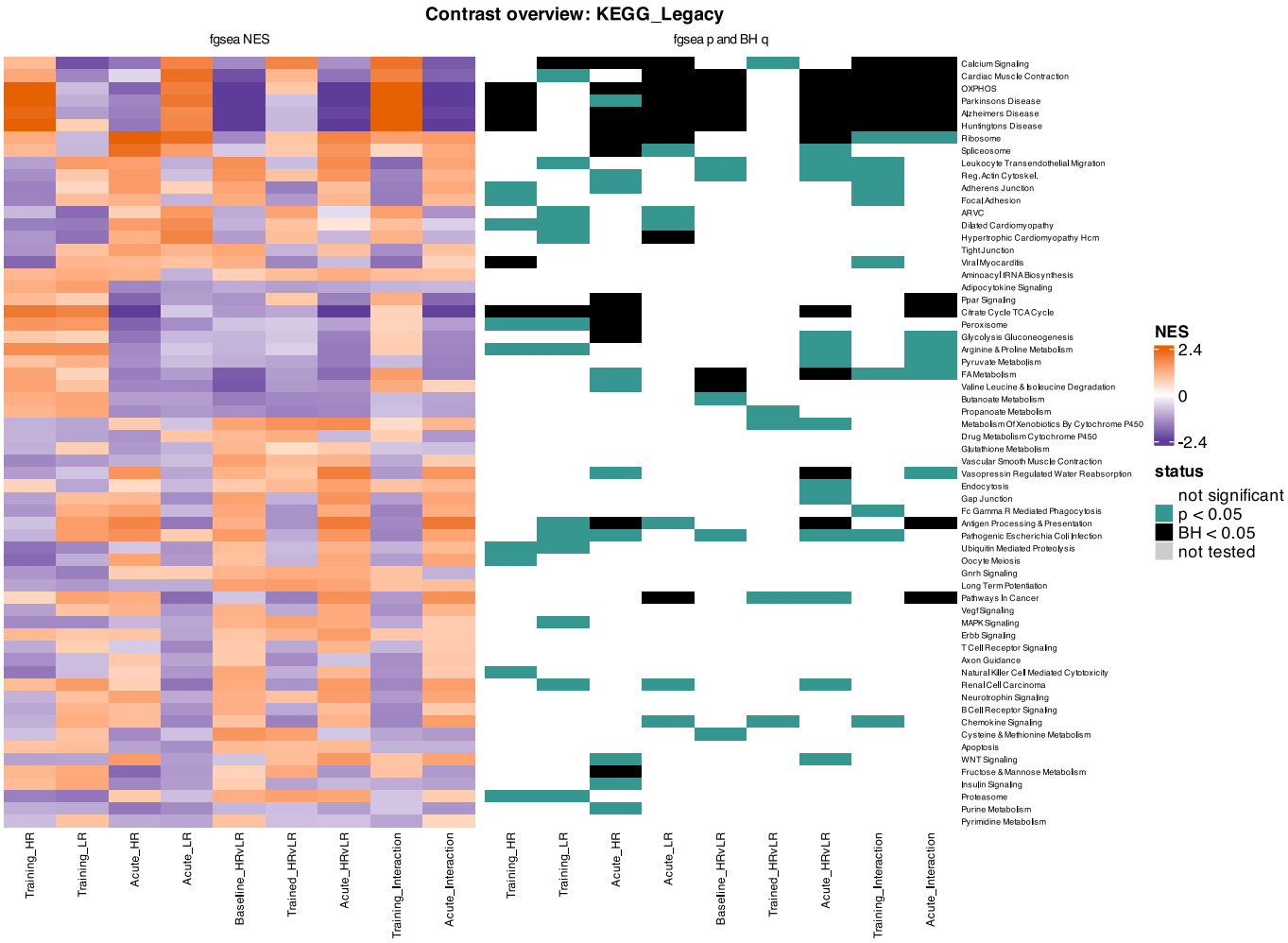


Figure 41: Every set across biopsies, then across the nine contrasts, by collection.



All 62 sets by 45 biopsies. Centred singscore; colours stop at 0.053.

Figure 42: Every set across biopsies, then across the nine contrasts, by collection.



fgsea NES; status: p and BH q within contrast, collections pooled. Colours stop at 2.4.

Figure 43: Every set across biopsies, then across the nine contrasts, by collection.

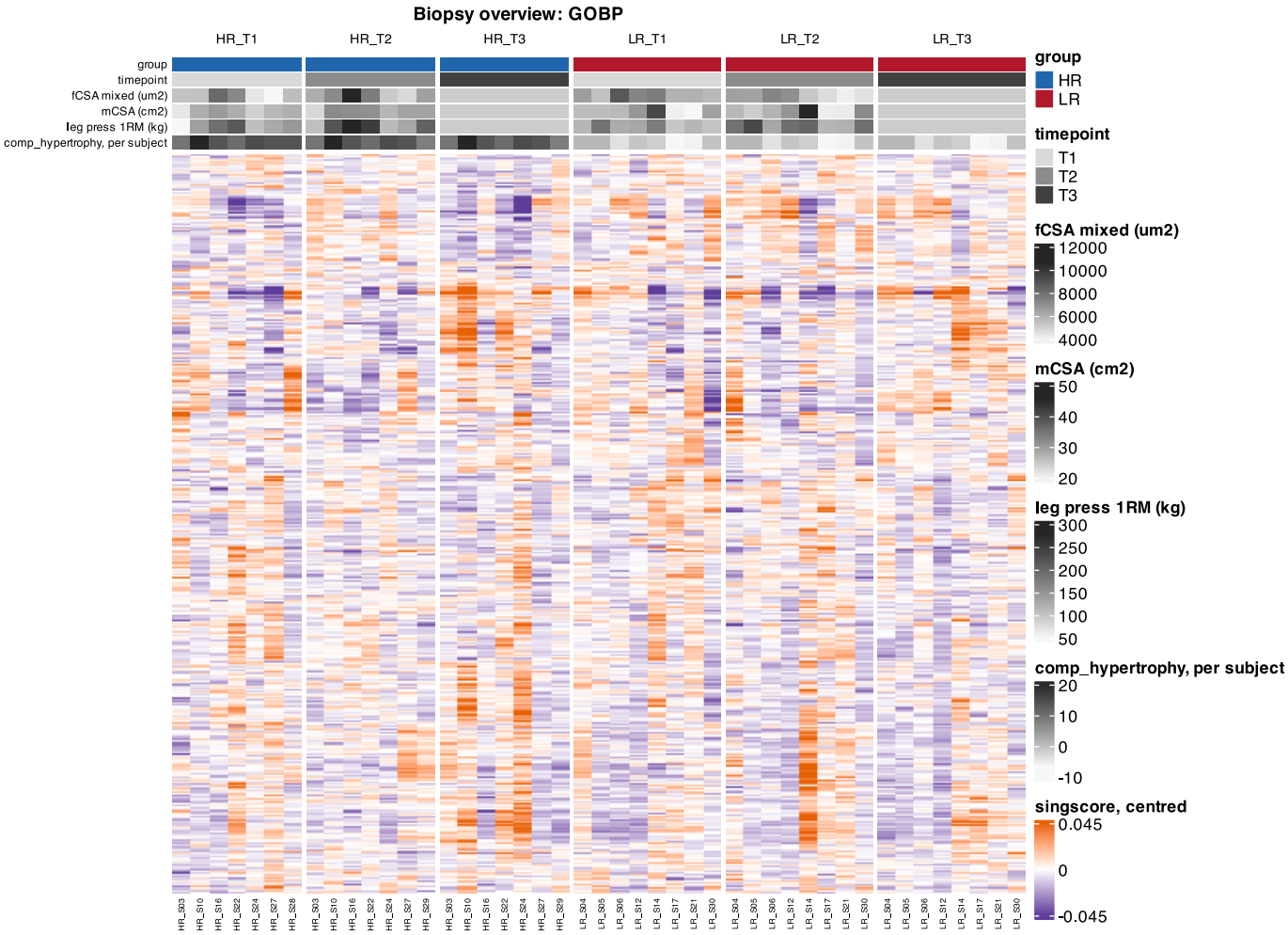


Figure 44: Every set across biopsies, then across the nine contrasts, by collection.

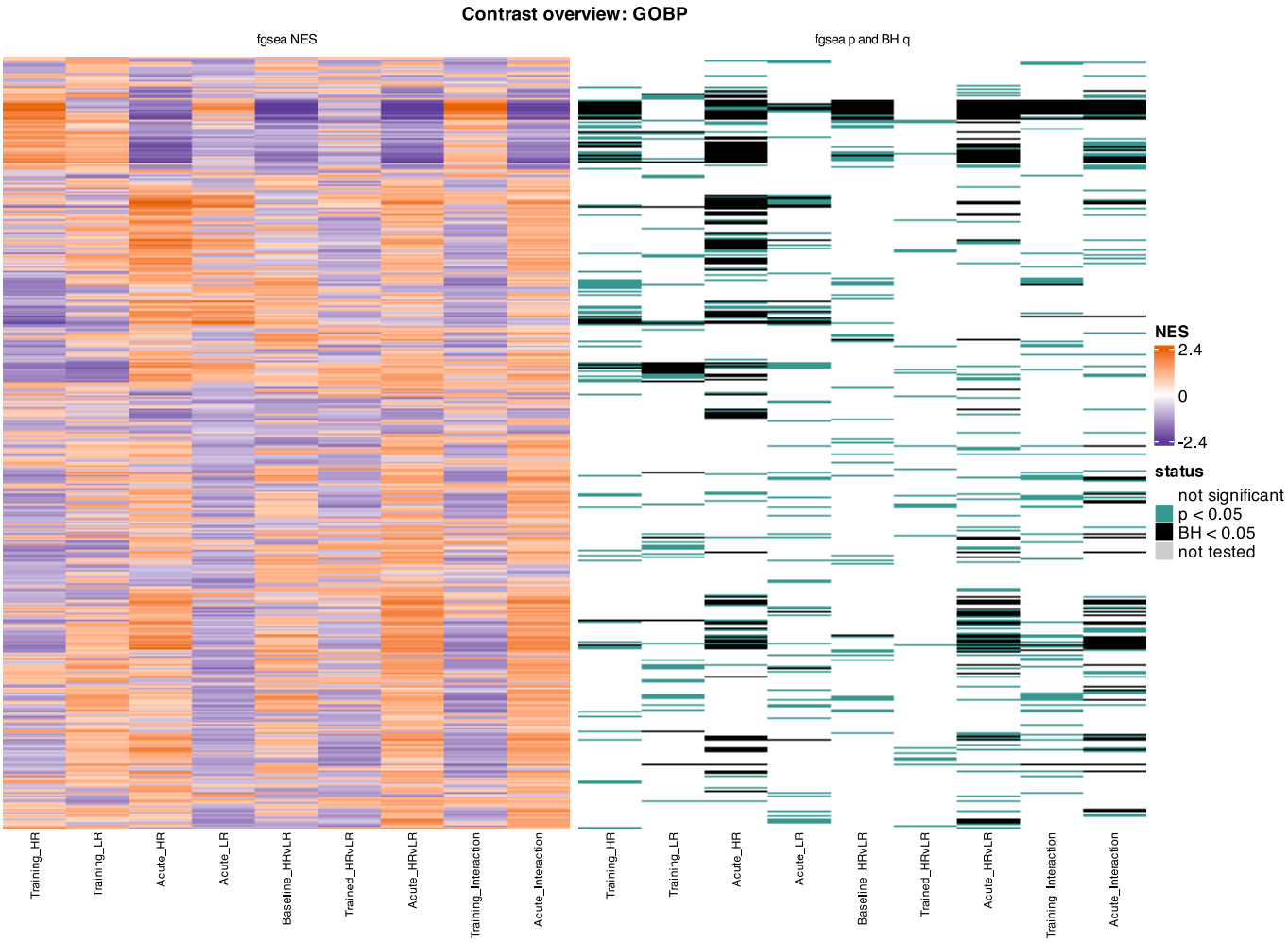
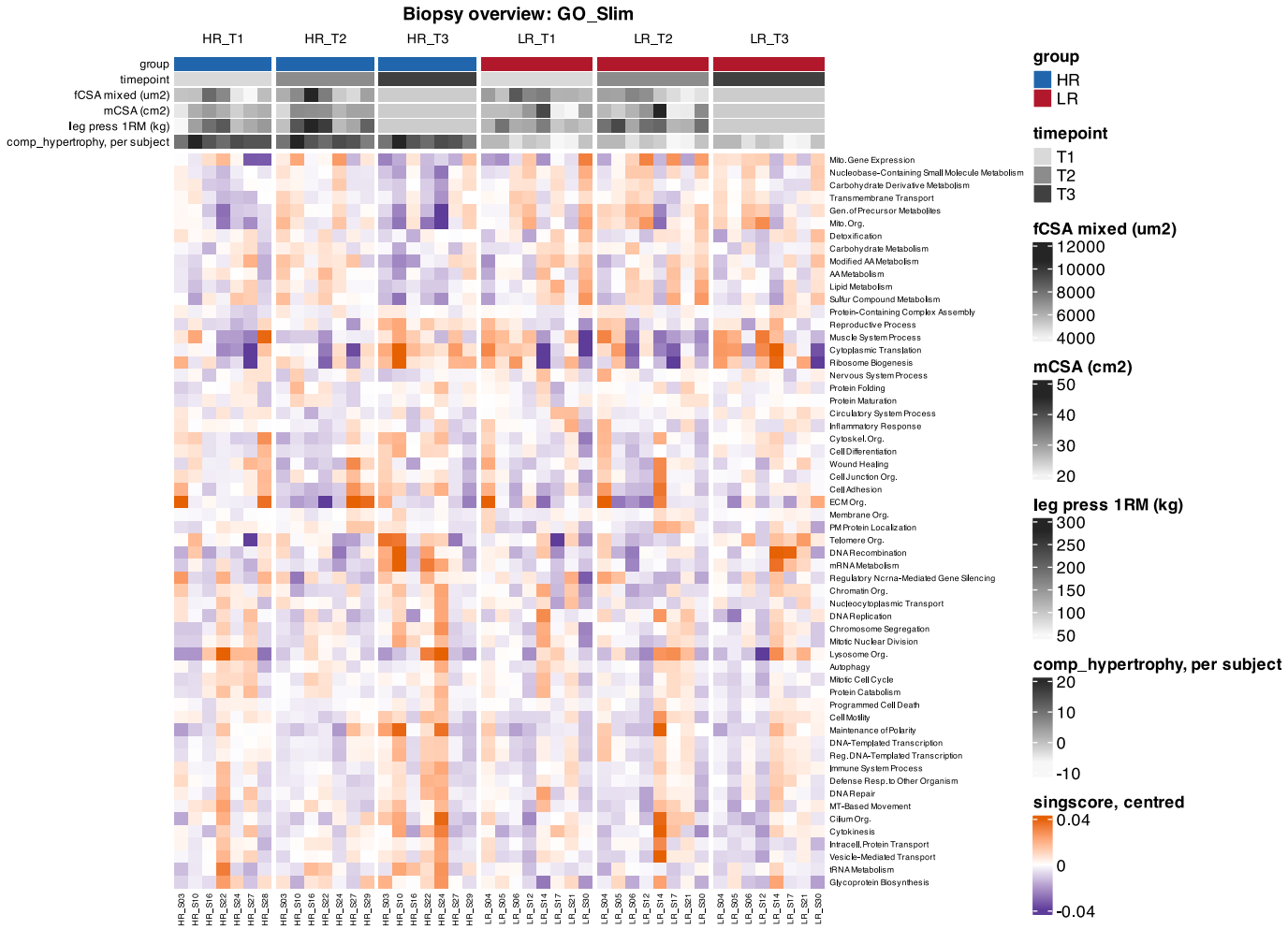


Figure 45: Every set across biopsies, then across the nine contrasts, by collection.



All 58 sets by 45 biopsies. Centred singscore; colours stop at 0.04.

Figure 46: Every set across biopsies, then across the nine contrasts, by collection.

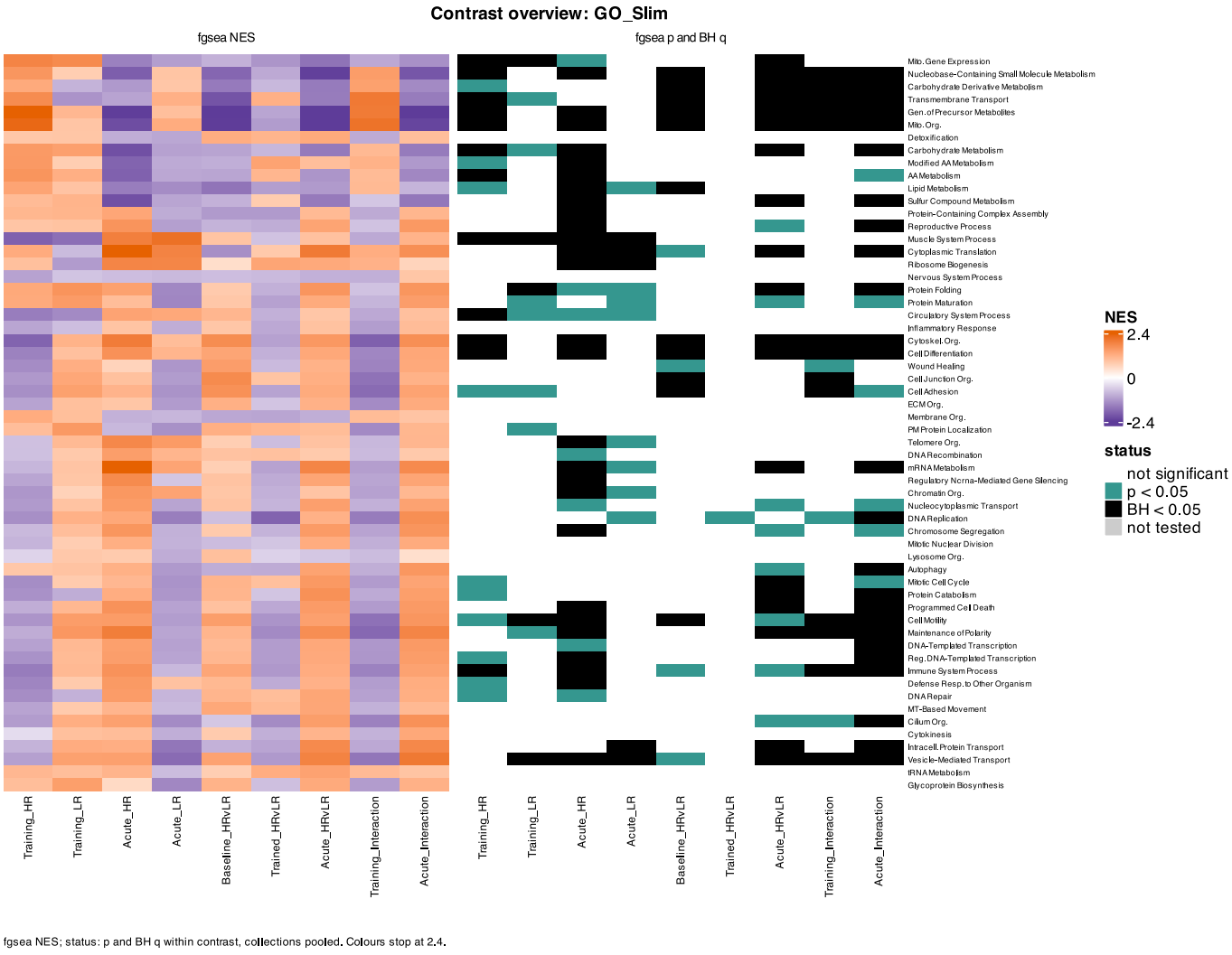


Figure 47: Every set across biopsies, then across the nine contrasts, by collection.

Validation

```
stopifnot(
  nrow(filter(set_tests, method == "fry")) == length(sets) * length(contrast_names),
  all(set_tests$fdr >= set_tests$p, na.rm = TRUE),
  all(filter(set_tests, method == "fgsea")$set_id %in% names(sets))
)
```

Summary

```
fgsea_calls <- filter(set_summary, contrast != d$baseline_reference)$n_fdr_fgsea
```

1378 sets were tested per contrast. fgsea calls 0 to 299 sets at FDR < 0.05 across the contrasts other than the baseline reference, which calls 70. fry calls 0 set-contrast pairs at FDR < 0.05 across all nine.

Save

`03_set_tests.rds` holds every protein result and every set test; the figures and the classification step read it.

```
out <- here(data_dir)
dir.create(out, recursive = TRUE, showWarnings = FALSE)
saveRDS(
  list(protein_results = protein_results, set_tests = set_tests),
  file.path(out, "03_set_tests.rds"),
  compress = "xz"
)
write_workbook(
  list(
    set_summary = set_summary,
    significant = set_tests |>
      filter(fdr < 0.05) |>
      mutate(leading_edge = map_chr(leading_edge, paste, collapse = ";")),
    protein_summary = protein_summary,
    fry_correlation = tibble(matrix = "limpa", correlation = correlation)
  ),
  c(
    "Sets tested and called per contrast by method, and fgsea collapse survivors.",
    "Every set at FDR < 0.05, either method. The full table is set_tests in set_tests.rds.",
    "Proteins tested and at BH < 0.05 per contrast.",
    "Within-subject correlation fry used, estimated on limpa's matrix."
  ),
  file.path(out, "03_set_contrasts.xlsx"),
  inputs, paths
)
```